Lab Work 1: Introduction to SQL Server and SSMS

Objective

To understand the fundamentals of SQL Server and SQL Server Management Studio (SSMS), create a database and table, insert sample insurance business data, and perform basic SQL queries for business intelligence analysis.

Theory

- **SQL Server:** Microsoft SQL Server is a Relational Database Management System (RDBMS) used to store, manage, and retrieve data efficiently. It supports Structured Query Language (SQL) for database operations and is widely used in enterprise applications.
- **SQL Server Management Studio (SSMS):** SQL Server Management Studio (SSMS) is a graphical user interface used to manage SQL Server databases. It provides tools for creating databases, designing tables, executing queries, and administering database servers.
- **Database and Tables:** A database is an organized collection of data. Tables are used to store data in rows and columns. Each table contains fields (columns) that define the type of information stored.
- **SQL Queries:** SQL queries are commands used to interact with databases. Common queries include:
  - SELECT: Retrieves data from a table.
  - WHERE: Filters records based on conditions.
  - ORDER BY: Sorts data in ascending or descending order.
  - GROUP BY: Groups records for aggregation and analysis.

Procedure

Step 1: Install SQL Server Express and SSMS

- Download SQL Server Express from Microsoft's official website.
- Run the installer and select the Basic installation option.
- Complete the installation process.
- Download SQL Server Management Studio (SSMS).
- Install SSMS using the default settings.

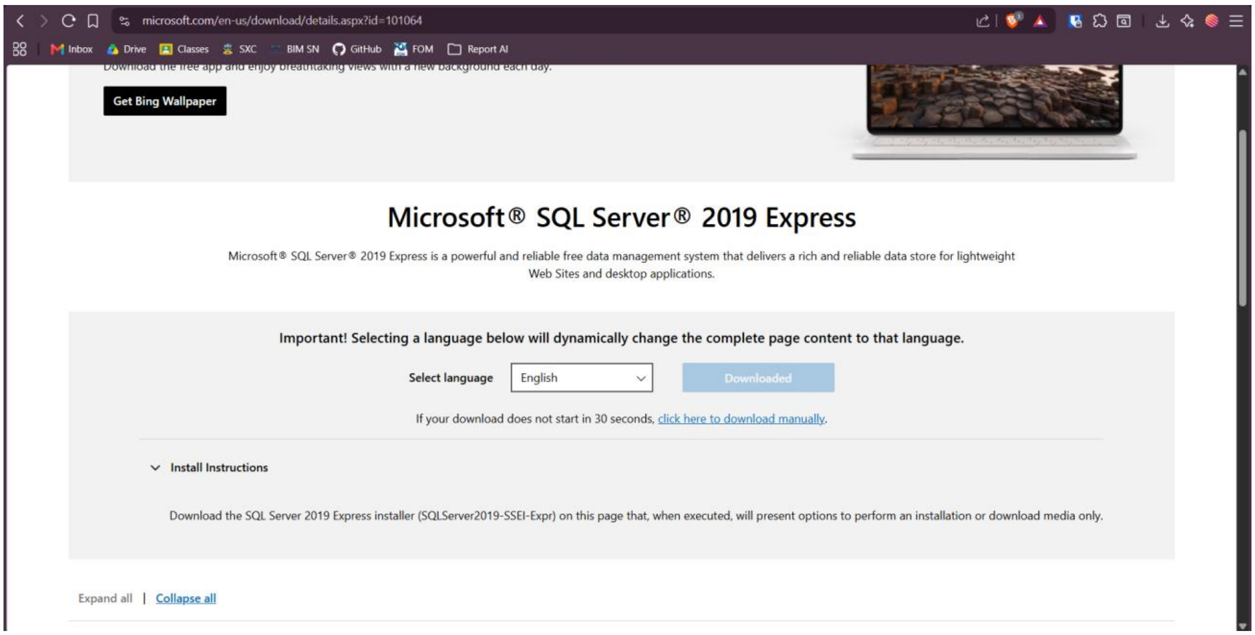


Figure 1.1 Download the SQL Server 2019

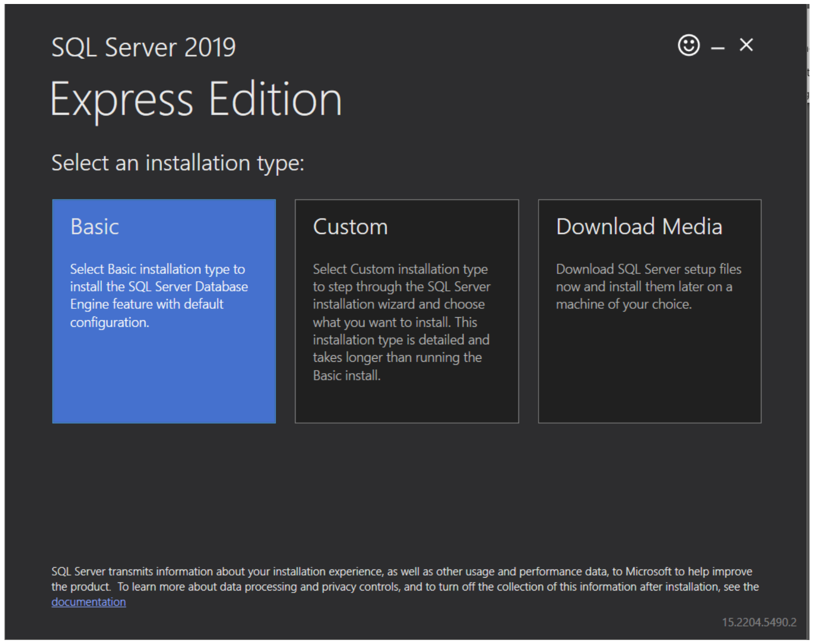


Figure 1.2 Selection of basic installation option

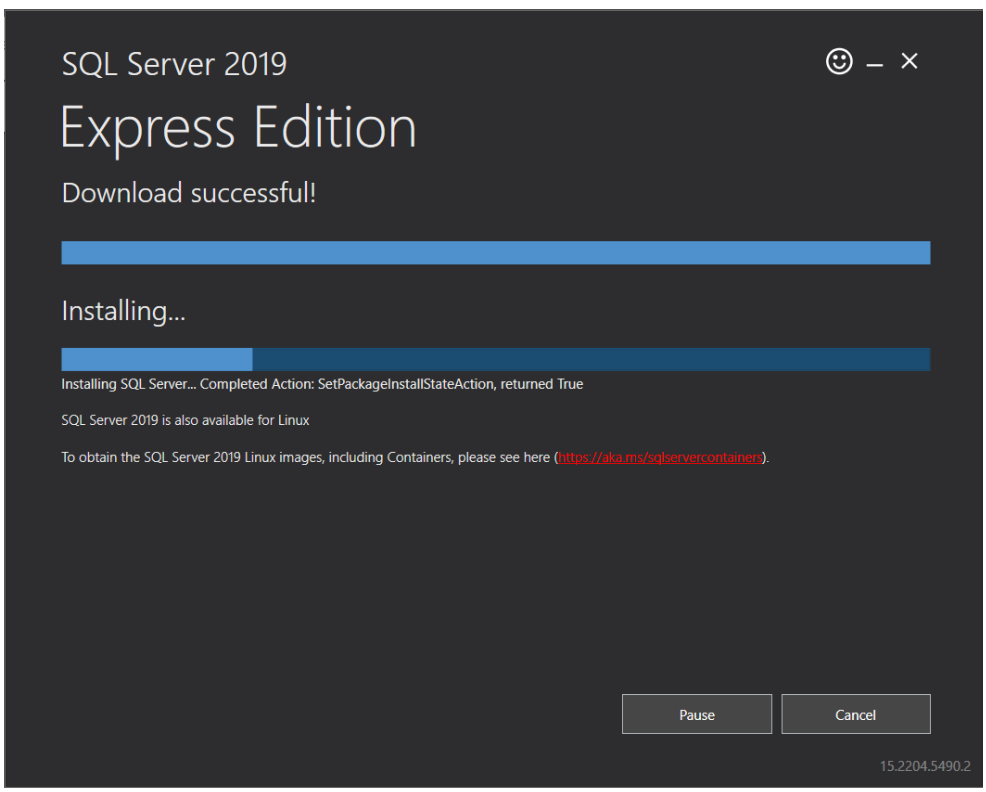


Figure 1.3 Installing process

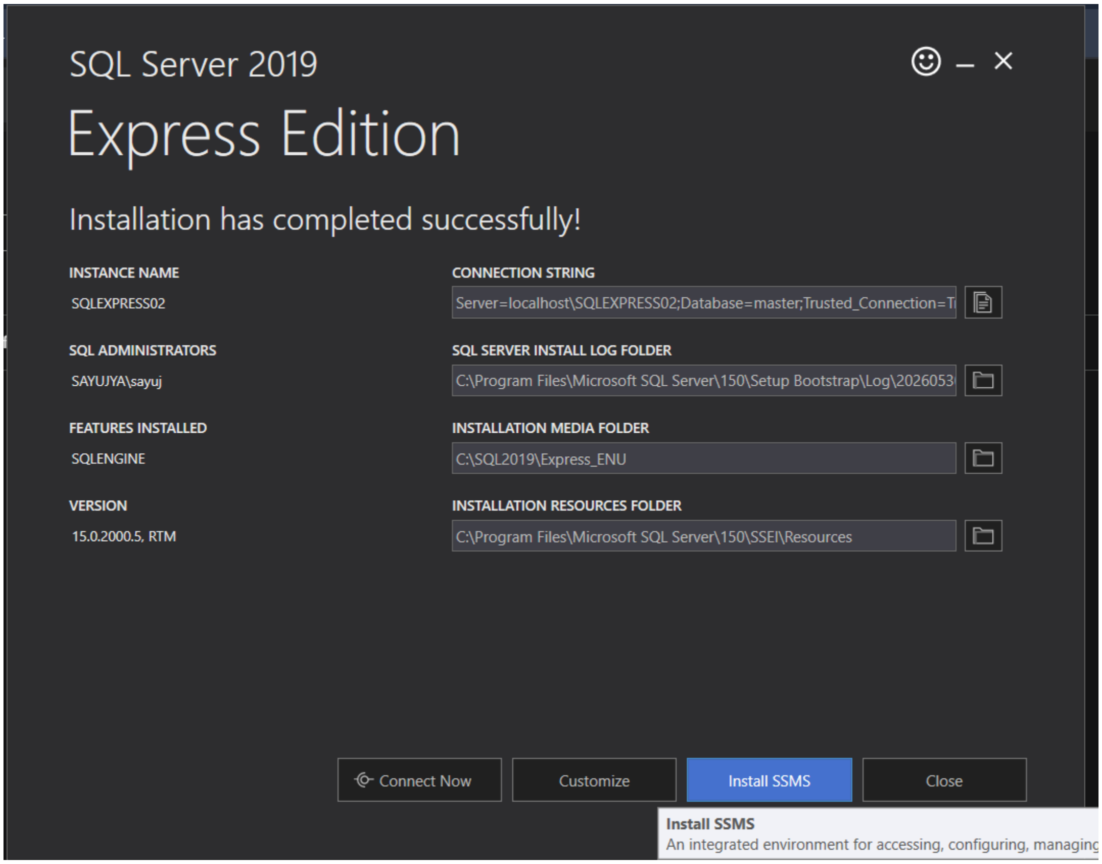


Figure 1.4 Option to download the SSMS

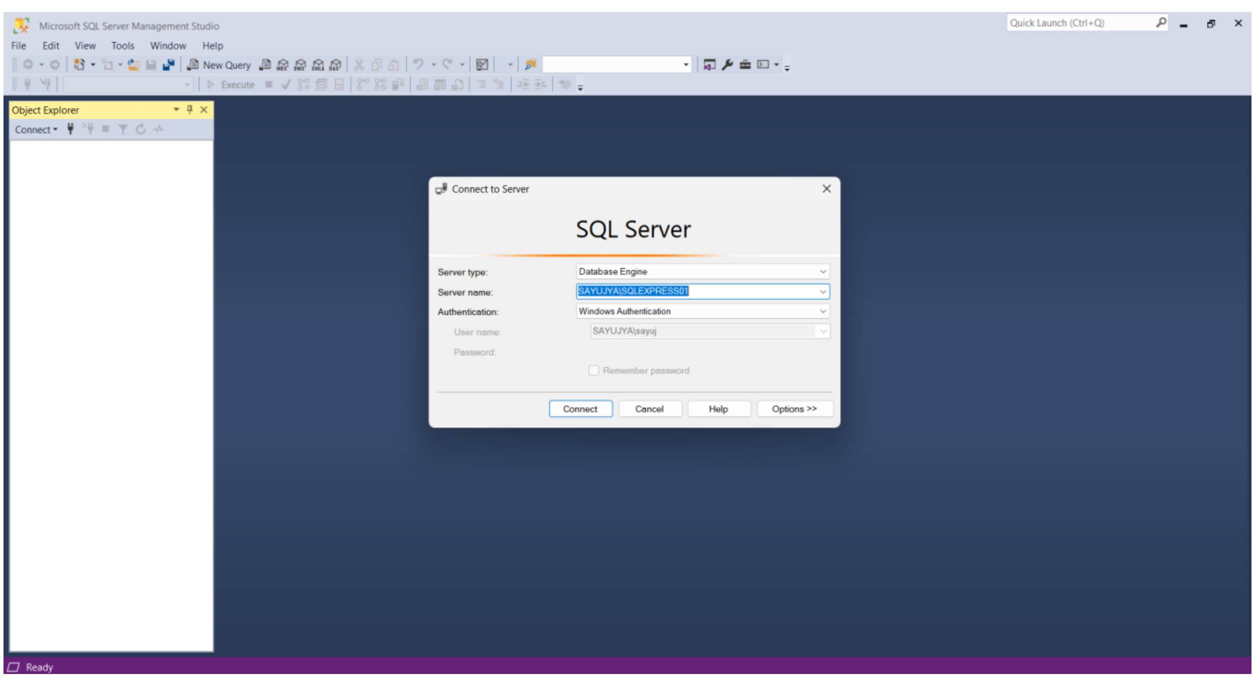


Figure 1.5 Successful installation of SSMS 19

Step 2: Connect to SQL Server Instance

- Open SQL Server Management Studio.
- In the Connect to Server window:
- Server Type: Database Engine
- Authentication: Windows Authentication
- Click Connect.

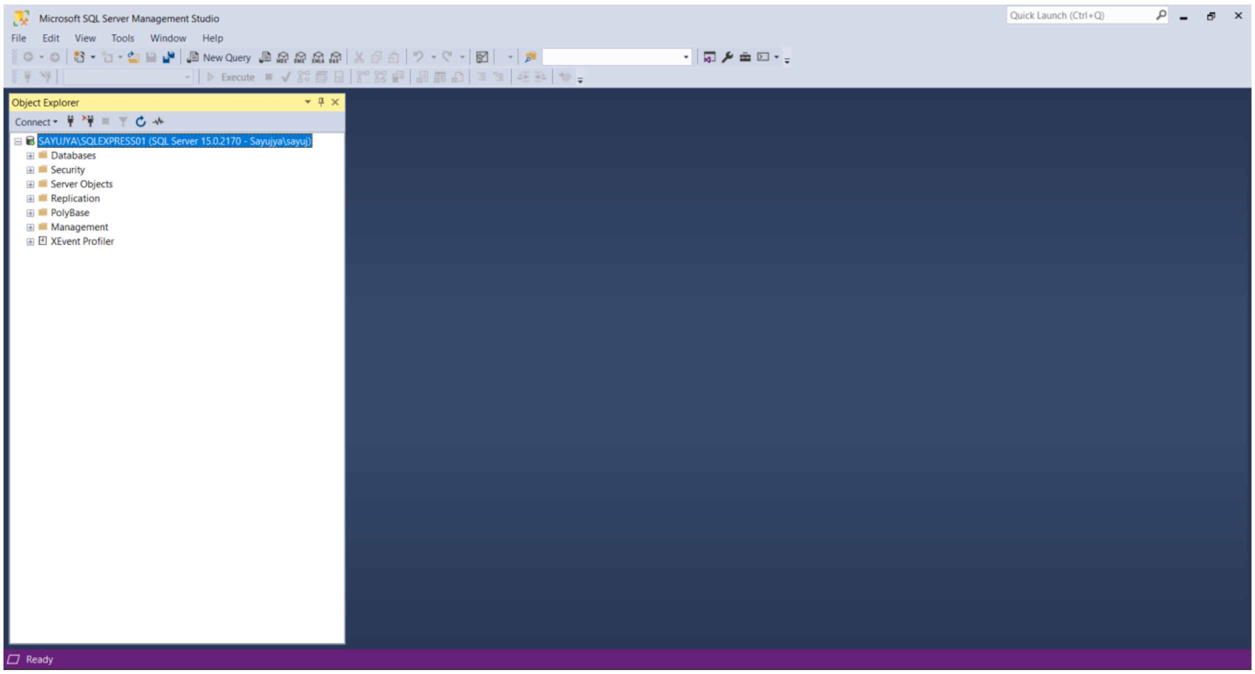


Figure 1.6 After successful connection

Step 3: Create Database

- Execute the following SQL command: `CREATE DATABASE Insurance_LabDB;`
- Refresh the Databases folder to verify creation.

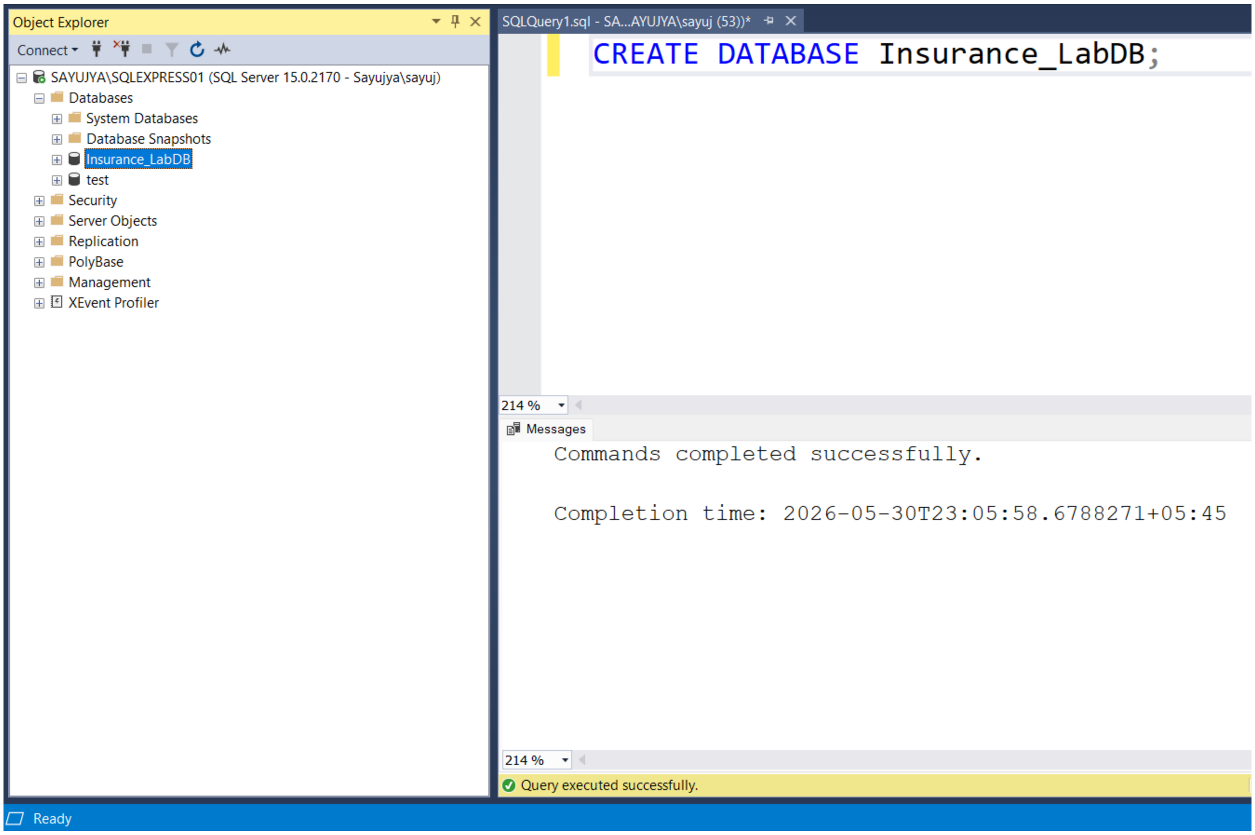


Figure 1.7 Successful creation of DB

Step 4: Create InsuranceBusiness Table

- Execute the following SQL statement:

```
USE Insurance_LabDB;

CREATE TABLE InsuranceBusiness (
    RecordID INT PRIMARY KEY,
    MonthYear VARCHAR(20),
    InsuranceCategory VARCHAR(50),
    PoliciesIssued INT,
    TotalPremium DECIMAL(10,2),
    SumInsured DECIMAL(10,2)
);
```

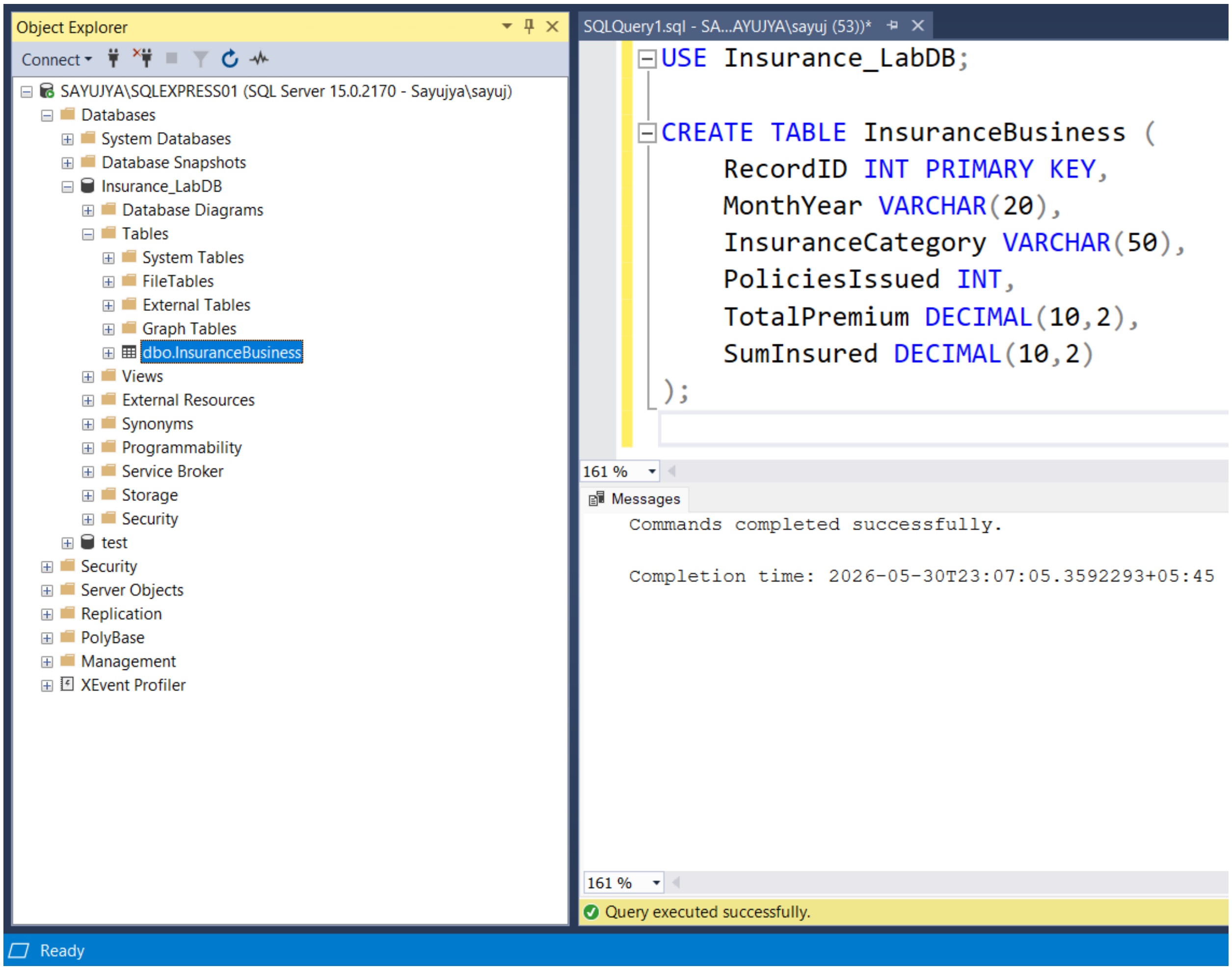


Figure 1.8 Successful creation of table

Step 5: Insert Sample Data

Execute the following SQL statement:

```
INSERT INTO InsuranceBusiness
VALUES
(1, 'Jan-2024', 'Non-Life', 1200, 450.50, 15000.00),
(2, 'Feb-2024', 'Micro Non-Life', 800, 220.75, 9000.00),
(3, 'Mar-2024', 'Non-Life', 1350, 500.25, 18000.00);
```

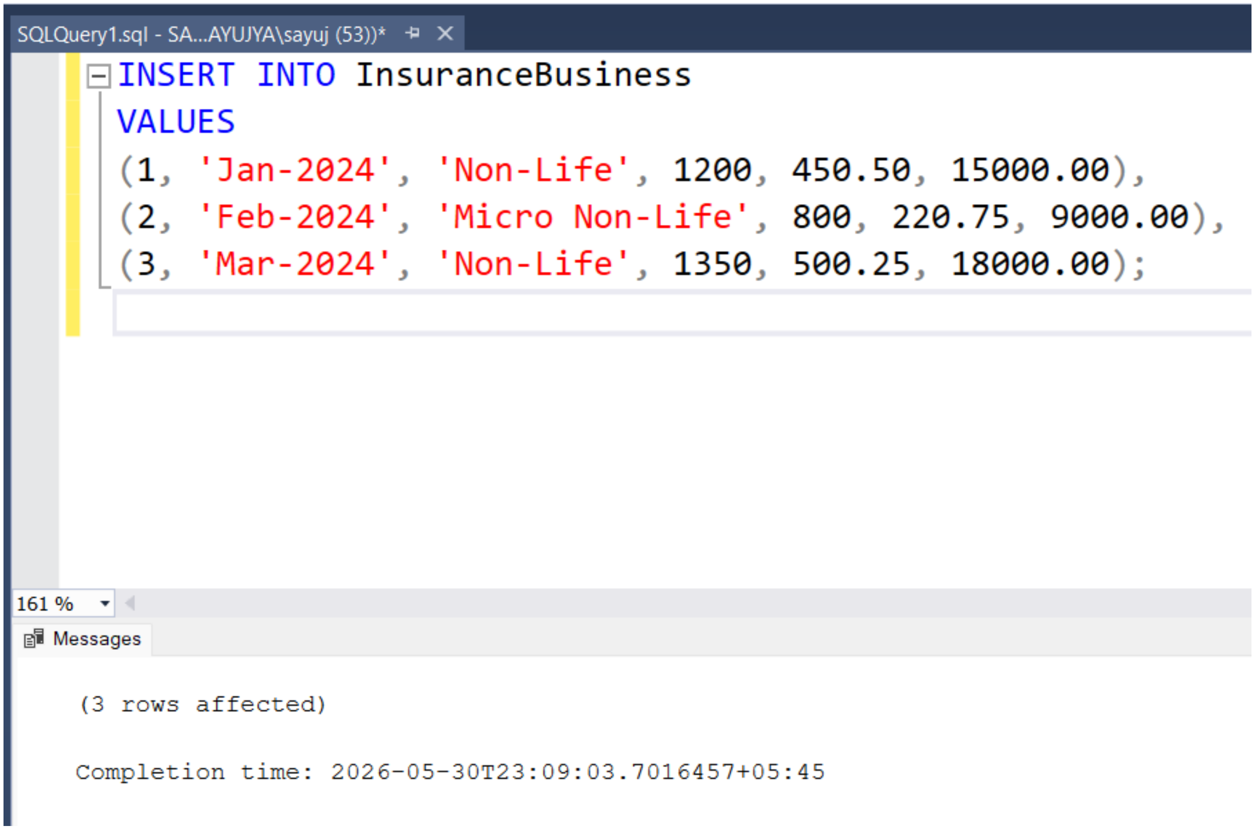
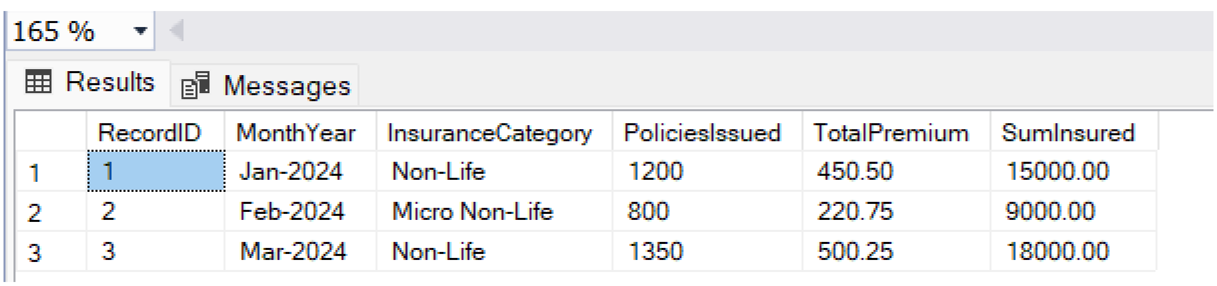


Figure 1.9 Successful insertion of sample data

Step 6: Execute Basic Queries

a. Display All Records

```
SELECT * FROM InsuranceBusiness;
```



	RecordID	MonthYear	InsuranceCategory	PoliciesIssued	TotalPremium	SumInsured
1	1	Jan-2024	Non-Life	1200	450.50	15000.00
2	2	Feb-2024	Micro Non-Life	800	220.75	9000.00
3	3	Mar-2024	Non-Life	1350	500.25	18000.00

Figure 1.10 All records output

b. Filter Records Using WHERE Clause

```
SELECT *
FROM InsuranceBusiness
WHERE InsuranceCategory = 'Non-Life';
```

	RecordID	MonthYear	InsuranceCategory	PoliciesIssued	TotalPremium	SumInsured
1	1	Jan-2024	Non-Life	1200	450.50	15000.00
2	3	Mar-2024	Non-Life	1350	500.25	18000.00

Figure 1.11 Output for WHERE clause

c. Sort Records Using ORDER BY

```
SELECT *
FROM InsuranceBusiness
ORDER BY MonthYear;
```

	RecordID	MonthYear	InsuranceCategory	PoliciesIssued	TotalPremium	SumInsured
1	2	Feb-2024	Micro Non-Life	800	220.75	9000.00
2	1	Jan-2024	Non-Life	1200	450.50	15000.00
3	3	Mar-2024	Non-Life	1350	500.25	18000.00

Figure 1.12 Output of ORDER BY query

d. Aggregate Data Using GROUP BY

```
SELECT InsuranceCategory,
       SUM(TotalPremium) AS TotalPremium
FROM InsuranceBusiness
GROUP BY InsuranceCategory;
```

	InsuranceCategory	TotalPremium
1	Micro Non-Life	220.75
2	Non-Life	950.75

Figure 1.13 Output of Aggregation query

Result

A database named Insurance_LabDB was successfully created using SQL Server Management Studio. The InsuranceBusiness table was created, sample insurance data was inserted, and SQL queries such as SELECT, WHERE, ORDER BY, and GROUP BY were executed successfully to analyze business data.

Conclusion

This lab provided practical experience with SQL Server and SSMS. It demonstrated how to create databases and tables, insert records, and retrieve meaningful information using SQL queries. These skills form the foundation for business intelligence and data analysis applications.

Lab Work 2: Data Warehousing and ETL using SQL Server (Nepal Insurance Data)

Objective

To understand data warehousing concepts and perform a real-world ETL (Extract, Transform, Load) process using monthly Non-Life and Micro Non-Life Insurance Business data published by the Nepal Insurance Authority.

Theory

- **Data Warehousing:** A Data Warehouse is a centralized repository designed to store integrated, historical, and subject-oriented data from multiple sources for reporting and analytical purposes. Unlike operational databases, data warehouses are optimized for querying and decision-making rather than transaction processing.
    - Characteristics of a Data Warehouse
        - Subject-Oriented
        - Integrated
        - Time-Variant
        - Non-Volatile
    - Benefits
        - Supports business intelligence and analytics
        - Enables trend analysis and forecasting
        - Provides a consolidated view of organizational data
- **ETL (Extract, Transform, Load):** ETL is the process used to collect data from source systems, transform it into a suitable format, and load it into a data warehouse.
    - **Extract:** Data is collected from source systems such as Excel files, databases, APIs, or applications.
    - **Transform:** The extracted data is cleaned and converted into a standardized format by:
        - Removing inconsistencies
        - Converting data types
        - Handling missing values

        - Renaming columns
        - Standardizing date formats
    - **Load:** The transformed data is loaded into the data warehouse for reporting and analysis.
- **OLTP (Online Transaction Processing):** OLTP systems are designed for day-to-day business operations and transactions. Characteristics:
    - Fast inserts and updates
    - Highly normalized structure
    - Supports operational activities
- **Data Warehouse (DW):** Data warehouses are designed for analytical processing. Characteristics:
    - Optimized for reporting
    - Historical data storage
    - Dimensional modeling
- **Star Schema:** A Star Schema is a dimensional model consisting of one central fact table connected to multiple dimension tables.
    - **Fact Table:** Contains measurable business data.
    - **Dimension Tables:** Contain descriptive information.

Requirements

- Microsoft SQL Server Express
- SQL Server Management Studio (SSMS)
- Nepal Insurance Authority Monthly Insurance Business Excel Files
- Windows Operating System

Procedure

Step 1: Create OLTP and Data Warehouse Databases

- Open SQL Server Management Studio and execute the following SQL statements:

```
CREATE DATABASE Insurance_OLTP;

CREATE DATABASE Insurance_DW;
```

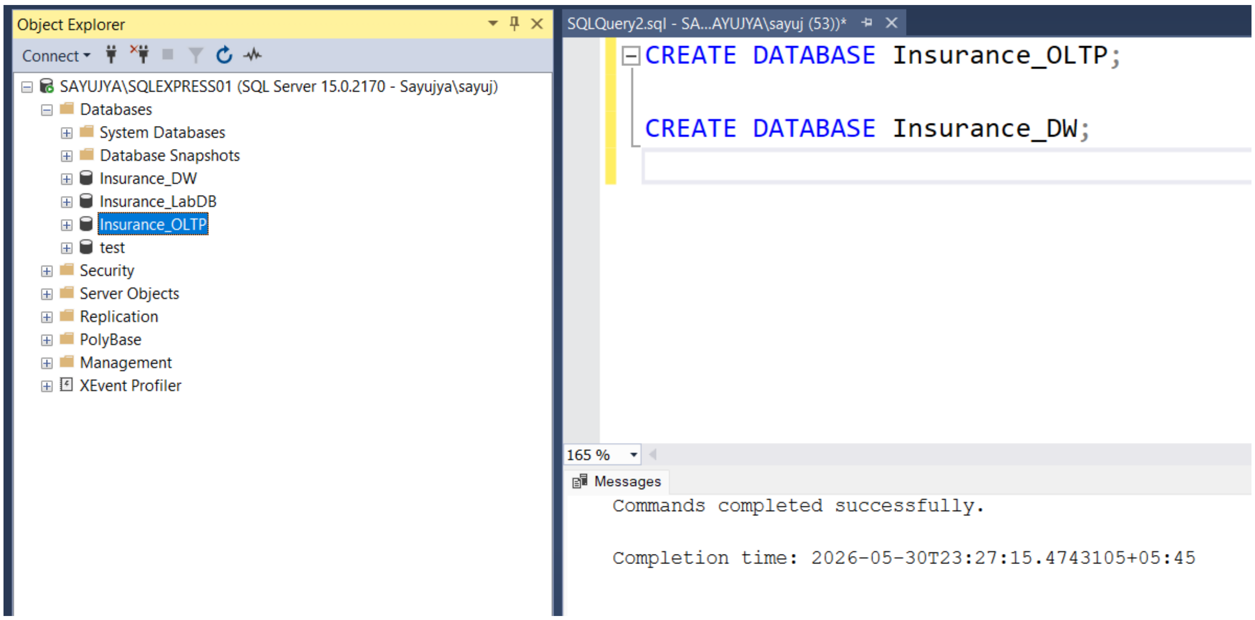


Figure 2.1 Successful creation of databases

Step 2: Download Monthly Insurance Data

- Open a web browser.
- Visit the Nepal Insurance Authority website.
- Navigate to the statistics or publications section.
- Download monthly Non-Life and Micro Non-Life Insurance Business Excel files.
- Save the files to a local folder.

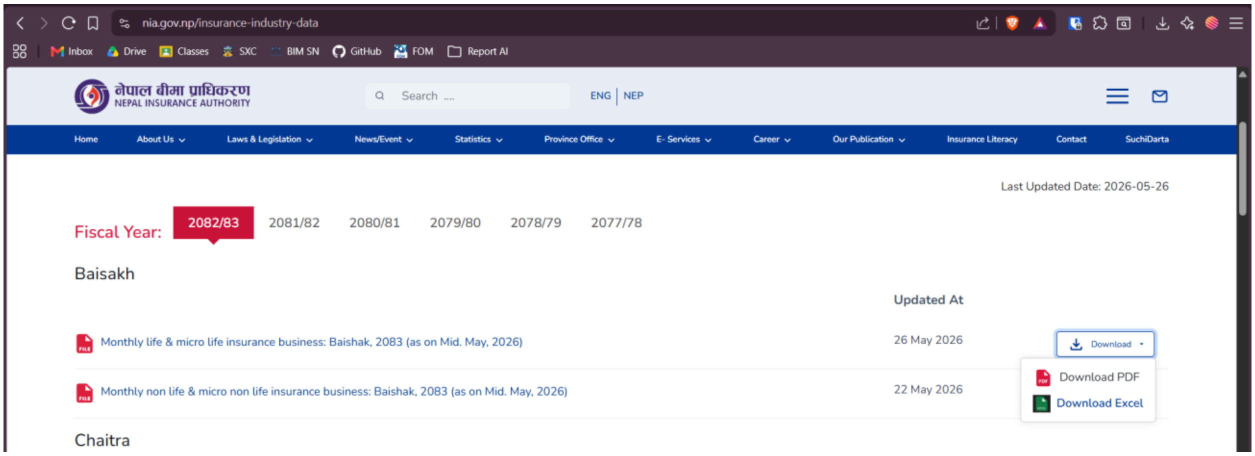
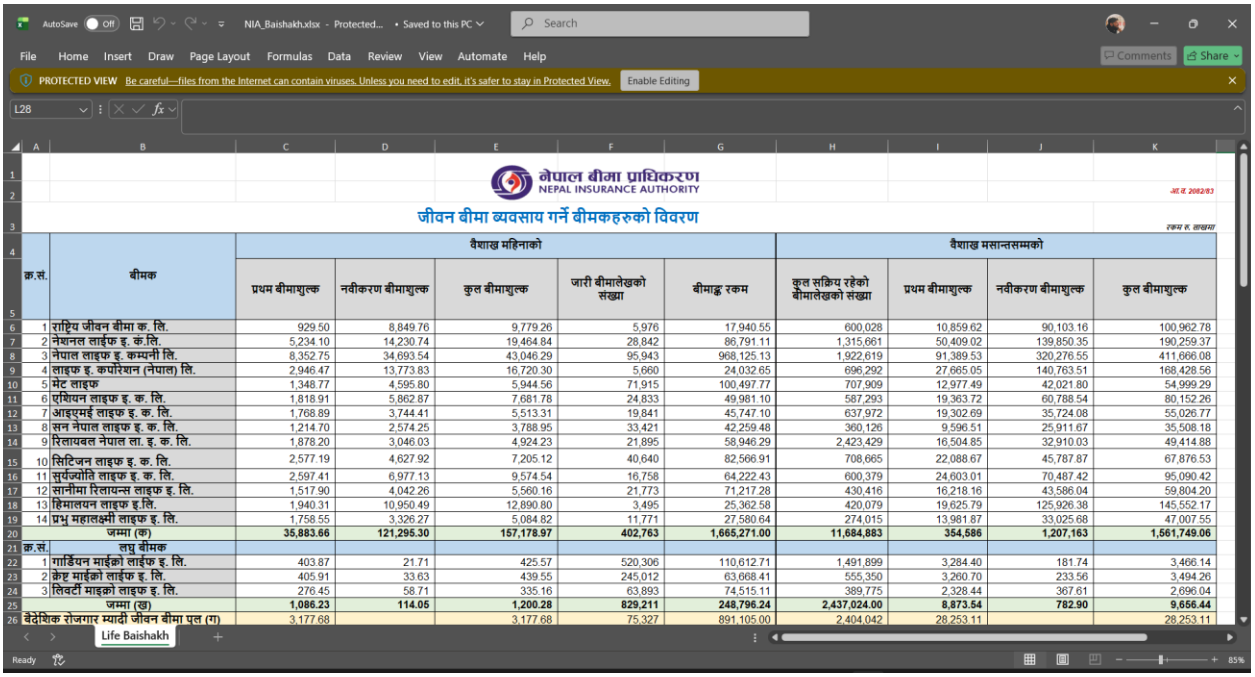


Figure 2.2 Option to download the file in excel



क्र.सं.	बीमक	वैशाख महिनाको					वैशाख मसान्तसम्मको			
		प्रथम बीमाशुल्क	नवीकरण बीमाशुल्क	कुल बीमाशुल्क	जारी बीमालेखको संख्या	बीमाङ्क रकम	कुल सक्रिय रहेको बीमालेखको संख्या	प्रथम बीमाशुल्क	नवीकरण बीमाशुल्क	कुल बीमाशुल्क
1	राष्ट्रिय जीवन बीमा क. लि.	929.50	8,849.76	9,779.26	5,976	17,940.55	600,028	10,859.62	90,103.16	100,962.78
2	नेशनल लाईफ इ. कं.लि.	5,234.10	14,230.74	19,464.84	28,842	86,791.11	1,315,661	50,409.02	139,850.35	190,259.37
3	नेपाल लाइफ इ. कम्पनी लि.	8,352.75	34,693.54	43,046.29	95,943	968,125.13	1,922,619	91,389.53	320,276.55	411,666.08
4	लाइफ इ. कर्पोरेशन (नेपाल) लि.	2,946.47	13,773.83	16,720.30	5,660	24,032.65	696,292	27,665.05	140,763.51	168,428.56
5	मेट लाइफ	1,348.77	4,595.80	5,944.56	71,915	100,497.77	707,909	12,977.49	42,021.80	54,999.29
6	एशियन लाइफ इ. क. लि.	1,818.91	5,862.87	7,681.78	24,833	49,981.10	587,293	19,363.72	60,788.54	80,152.26
7	आइएमई लाइफ इ. क. लि.	1,768.89	3,744.41	5,513.31	19,841	45,747.10	637,972	19,302.69	35,724.08	55,026.77
8	सन नेपाल लाइफ इ. क. लि.	1,214.70	2,574.25	3,788.95	33,421	42,259.48	360,126	9,596.51	25,911.67	35,508.18
9	रिलायबल नेपाल ला. इ. क. लि.	1,878.20	3,046.03	4,924.23	21,895	58,946.29	2,423,429	16,504.85	32,910.03	49,414.88
10	सिटिजन लाइफ इ. क. लि.	2,577.19	4,627.92	7,205.12	40,640	82,566.91	708,665	22,088.67	45,787.87	67,876.53
11	सुर्यज्योति लाइफ इ. क. लि.	2,597.41	6,977.13	9,574.54	16,758	64,222.43	600,379	24,603.01	70,487.42	95,090.42
12	सानीमा रिलायन्स लाइफ इ. लि.	1,517.90	4,042.26	5,560.16	21,773	71,217.28	430,416	16,218.16	43,586.04	59,804.20
13	हिमालयन लाइफ इ.लि.	1,940.31	10,950.49	12,890.80	3,495	25,362.58	420,079	19,625.79	125,926.38	145,552.17
14	प्रभु महालक्ष्मी लाइफ इ. लि.	1,758.55	3,326.27	5,084.82	11,771	27,580.64	274,015	13,981.87	33,025.68	47,007.55
	जम्मा (क)	35,883.66	121,295.30	157,178.97	402,763	1,665,271.00	11,684,883	354,586	1,207,163	1,561,749.06
क्र.सं.	लघु बीमक									
1	गार्डियन माईक्रो लाईफ इ. लि.	403.87	21.71	425.57	520,306	110,612.71	1,491,899	3,284.40	181.74	3,466.14
2	क्रेष्ट माईक्रो लाईफ इ. लि.	405.91	33.63	439.55	245,012	63,668.41	555,350	3,260.70	233.56	3,494.26
3	लिवर्टी माइक्रो लाइफ इ. लि.	276.45	58.71	335.16	63,893	74,515.11	389,775	2,328.44	367.61	2,696.04
	जम्मा (ख)	1,086.23	114.05	1,200.28	829,211	248,796.24	2,437,024.00	8,873.54	782.90	9,656.44
वैदेशिक रोजगार म्यादी जीवन बीमा पुल (ग)		3,177.68		3,177.68	75,327	891,105.00	2,404,042	28,253.11		28,253.11

Figure 2.3 Downloaded excel report

Step 3: Create Staging Table in Insurance_OLTP

- Switch to the Insurance_OLTP database and create a staging table.

```sql
USE Insurance_OLTP;

CREATE TABLE Stg_Monthly_Insurance_Business
(
    RecordID INT IDENTITY(1,1) PRIMARY KEY,
    MonthYear NVARCHAR(50),
    NonLifePoliciesIssued INT,
    NonLifeTotalPremium DECIMAL(18,2),
    NonLifeSumInsured DECIMAL(18,2),
    MicroPoliciesIssued INT,
    MicroTotalPremium DECIMAL(18,2)
);
```

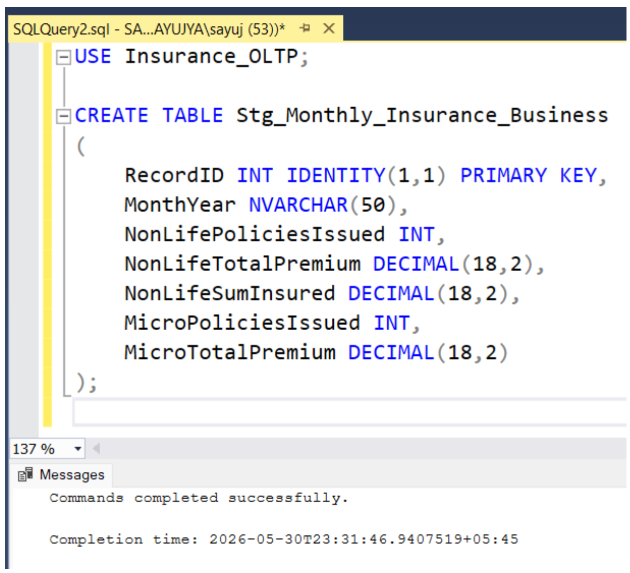


Figure 2.4 Successful creation of monthly insurance business table

Step 4: Import Excel Data into Staging Table

- Right-click the Insurance_OLTP database.
- Select Tasks > Import Data.
- Choose Microsoft Excel as the data source.
- Browse and select the downloaded Excel file.
- Select the destination database as Insurance_OLTP.
- Map Excel columns to the staging table.
- Complete the import process.

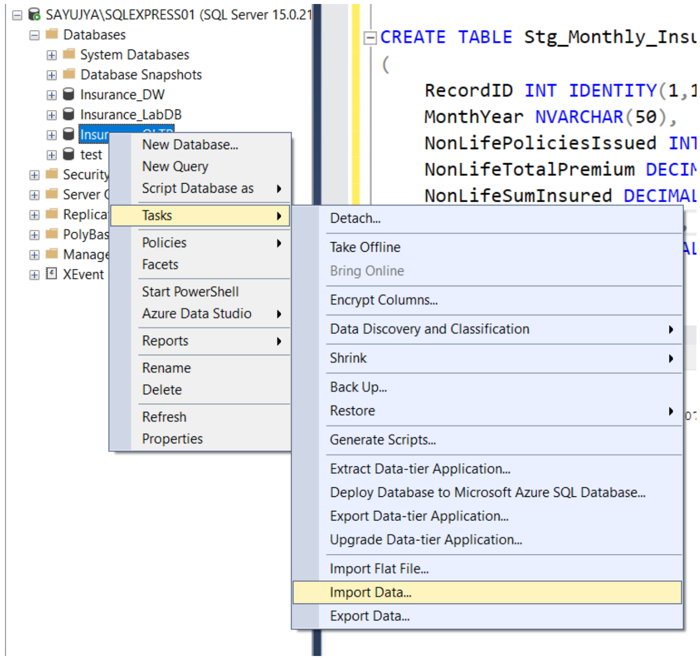


Figure 2.5 Selection of Import Data option

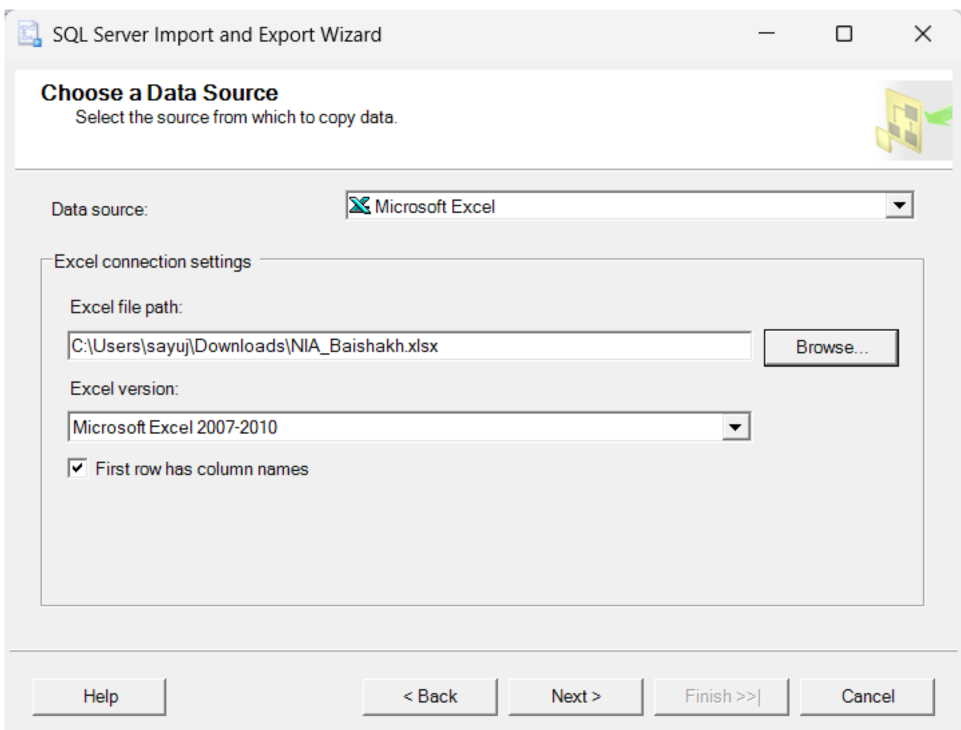


Figure 2.6 Selection of the excel file

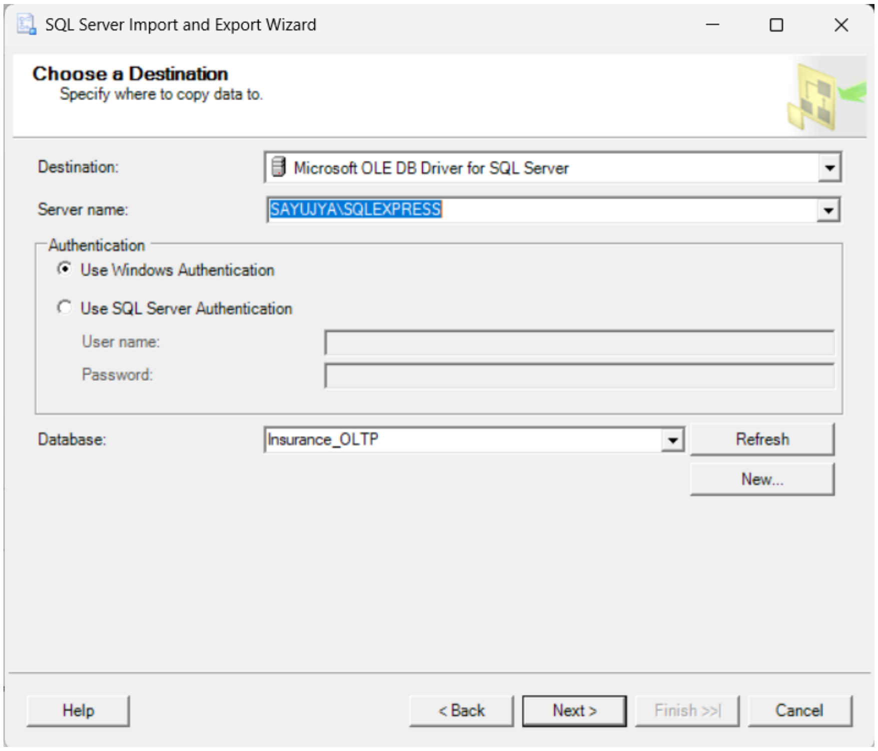


Figure 2.7 Selection of destination

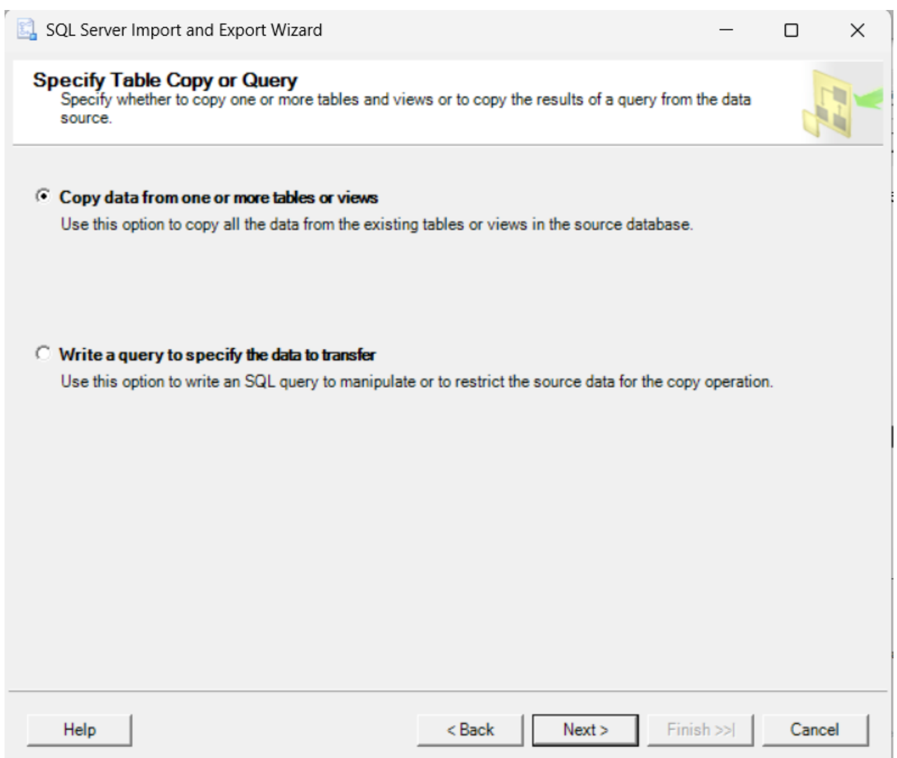


Figure 2.8 Selection of first option to copy data

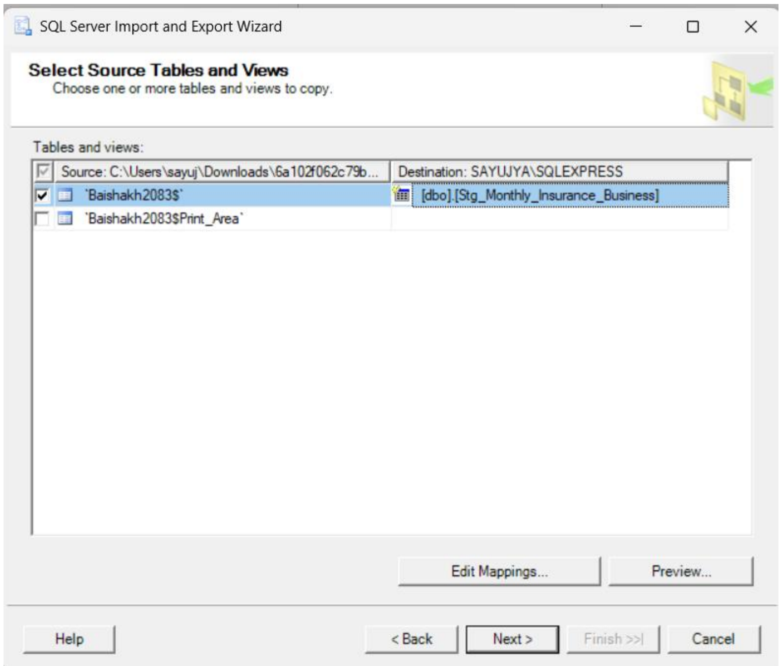


Figure 2.9 Selection of destination table

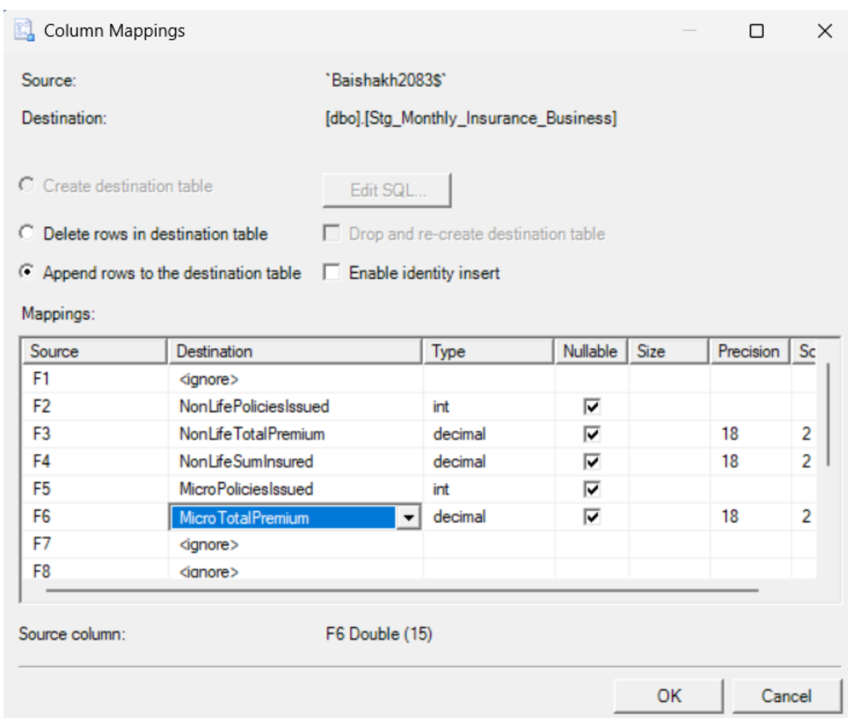


Figure 2.10 Mapping the fields to db columns

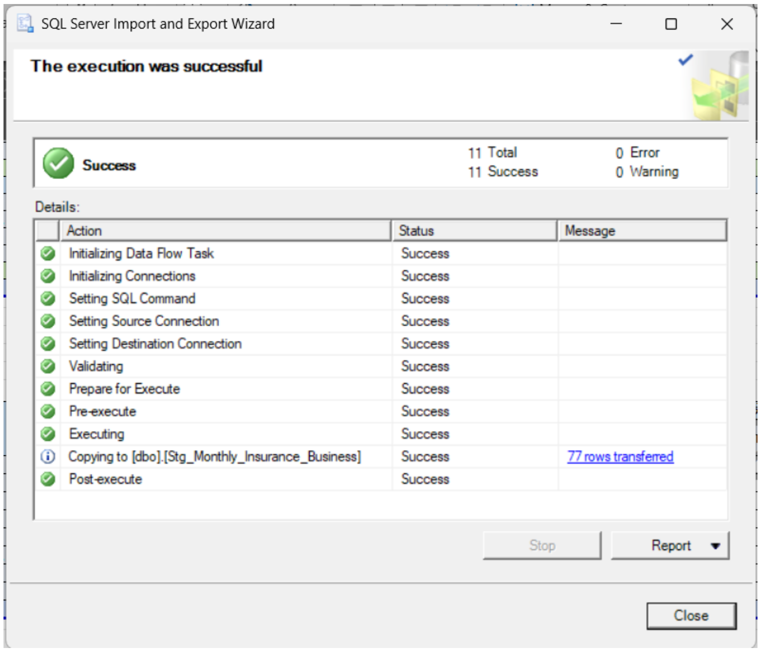


Figure 2.11 Successful transfer of the data

	RecordID	MonthYear	NonLifePoliciesIssued	NonLifeTotalPremium	NonLifeSumInsured	MicroPoliciesIssued	MicroTotalPremium
1	1	NULL	NULL	NULL	NULL	NULL	NULL
2	2	NULL	NULL	NULL	NULL	NULL	NULL
3	3	NULL	NULL	NULL	NULL	NULL	NULL
4	4	NULL	NULL	NULL	NULL	NULL	NULL
5	5	NULL	NULL	NULL	NULL	NULL	NULL
6	6	NULL	NULL	NULL	NULL	NULL	NULL
7	7	NULL	15142	1470.37	424778.92	162065	18157.31
8	8	NULL	4034	1600.08	433781.38	38245	25868.10
9	9	NULL	1824	388.64	3629473.14	25269	11885.67
10	10	NULL	2491	832.96	227223.62	22084	10645.89
11	11	NULL	26425	2894.58	729909.00	251704	30185.59
12	12	NULL	9615	1294.74	219018.13	92036	14976.45
13	13	NULL	29977	3909.64	1050576.41	268644	51922.63
14	14	NULL	18510	4330.33	559702.90	180403	34436.20
15	15	NULL	31306	6348.59	1612891.16	281173	43779.11
16	16	NULL	9875	2808.67	687165.18	104945	23628.52
17	17	NULL	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Figure 2.12 Final data view after transfer

Step 5: Perform Data Cleaning and Transformation

- **Fill MonthYear column with 01/2083**

```
UPDATE Stg_Monthly_Insurance_Business
SET MonthYear = '01/2083';
```

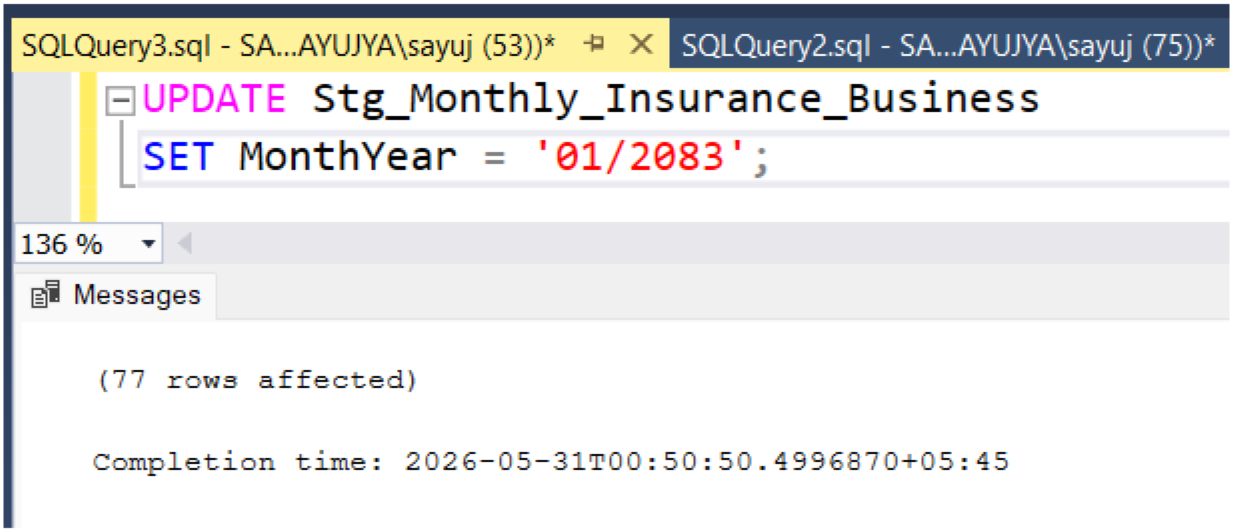


Figure 2.13 Query to update the MonthYear

- **Handle Missing Values**

Replace NULL values with default values.

```
UPDATE Stg_Monthly_Insurance_Business
SET NonLifePoliciesIssued = ISNULL(NonLifePoliciesIssued,0),
    NonLifeTotalPremium = ISNULL(NonLifeTotalPremium,0),
    NonLifeSumInsured = ISNULL(NonLifeSumInsured,0),
    MicroPoliciesIssued = ISNULL(MicroPoliciesIssued,0),
    MicroTotalPremium = ISNULL(MicroTotalPremium,0);
```

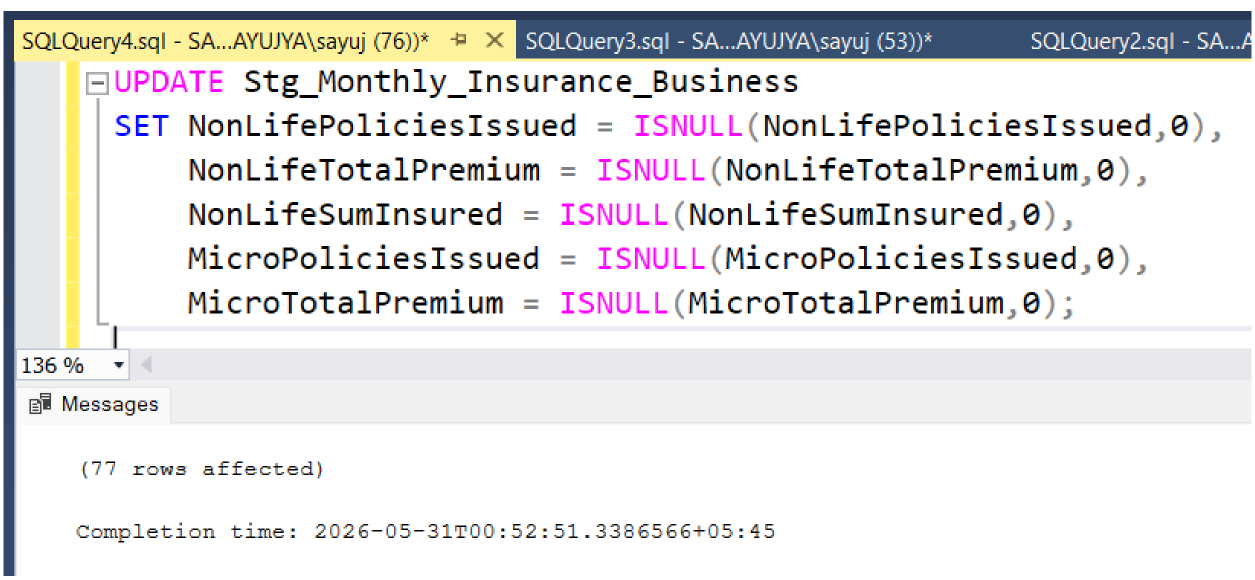


Figure 2.14 Query to remove any missing values

	RecordID	MonthYear	NonLifePoliciesIssued	NonLifeTotalPremium	NonLifeSumInsured	MicroPoliciesIssued	MicroTotalPremium
1	1	01/2083	0	0.00	0.00	0	0.00
2	2	01/2083	0	0.00	0.00	0	0.00
3	3	01/2083	0	0.00	0.00	0	0.00
4	4	01/2083	0	0.00	0.00	0	0.00
5	5	01/2083	0	0.00	0.00	0	0.00
6	6	01/2083	0	0.00	0.00	0	0.00
7	7	01/2083	15142	1470.37	424778.92	162065	18157.31
8	8	01/2083	4034	1600.08	433781.38	38245	25868.10
9	9	01/2083	1824	388.64	3629473.14	25269	11885.67
10	10	01/2083	2491	832.96	227223.62	22084	10645.89
11	11	01/2083	26425	2894.58	729909.00	251704	30185.59
12	12	01/2083	9615	1294.74	219018.13	92036	14976.45
13	13	01/2083	29977	3909.64	1050576.41	268644	51922.63
14	14	01/2083	18510	4330.33	559702.90	180403	34436.20
15	15	01/2083	31306	6348.59	1612891.16	281173	43779.11
16	16	01/2083	9875	2808.67	687165.18	104945	23628.52
17	17	01/2083	20196	3735.85	830337.03	225784	34945.14
18	18	01/2083	21375	4263.20	971478.00	223213	42586.86
19	19	01/2083	19141	4149.67	1096039.42	190594	35183.34
20	20	01/2083	31924	2118.33	809592.93	345503	24756.86
21	21	01/2083	241835	40145.73	13281967.28	2411662	402957.73

Figure 2.15 Final result after data cleaning

Step 6: Design the Star Schema in Insurance_DW

- Switch to the Insurance_DW database and create dimension and fact tables.

```
USE Insurance_DW;
CREATE TABLE DimInsuranceCategory
(
    CategoryID INT PRIMARY KEY,
    CategoryName VARCHAR(50)
);

CREATE TABLE DimDate
(
    DateID INT IDENTITY(1,1) PRIMARY KEY,
    MonthName VARCHAR(20),
    YearValue INT
);

CREATE TABLE FactInsuranceBusiness
(
    FactID INT IDENTITY(1,1) PRIMARY KEY,
```

```
    DateID INT,
    CategoryID INT,
    PoliciesCount INT,
    TotalPremium DECIMAL(18,2),
    SumInsured DECIMAL(18,2)
);
```

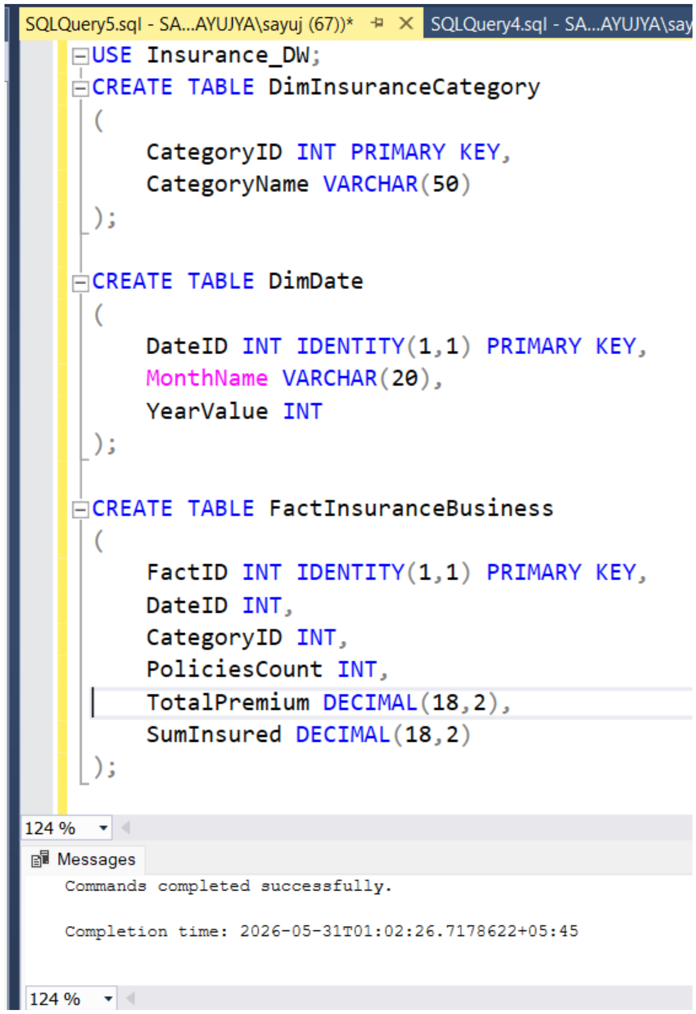


Figure 2.16 Successful creation of dimension and fact tables

Step 7: Populate Dimension Tables

- Insert Insurance Categories

```
INSERT INTO DimInsuranceCategory
VALUES
(1,'Non-Life'),
(2,'Micro Non-Life');
```

```
INSERT INTO DimDate
VALUES
('Baishakh', 2083)
```

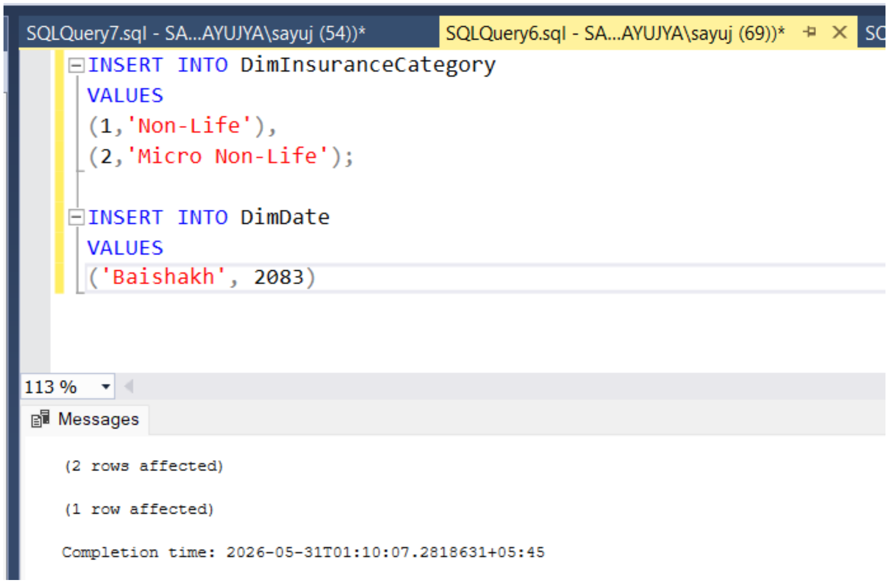


Figure 2.17 Successful Insertion of category

Step 8: Load Data into the Data Warehouse

Load transformed records from the staging table into the warehouse structure.

```
INSERT INTO FactInsuranceBusiness
(
    DateID,
    CategoryID,
    PoliciesCount,
    TotalPremium,
    SumInsured
)
SELECT
    1,
    1,
    NonLifePoliciesIssued,
    NonLifeTotalPremium,
    NonLifeSumInsured
FROM Insurance_OLTP.dbo.Stg_Monthly_Insurance_Business;
```

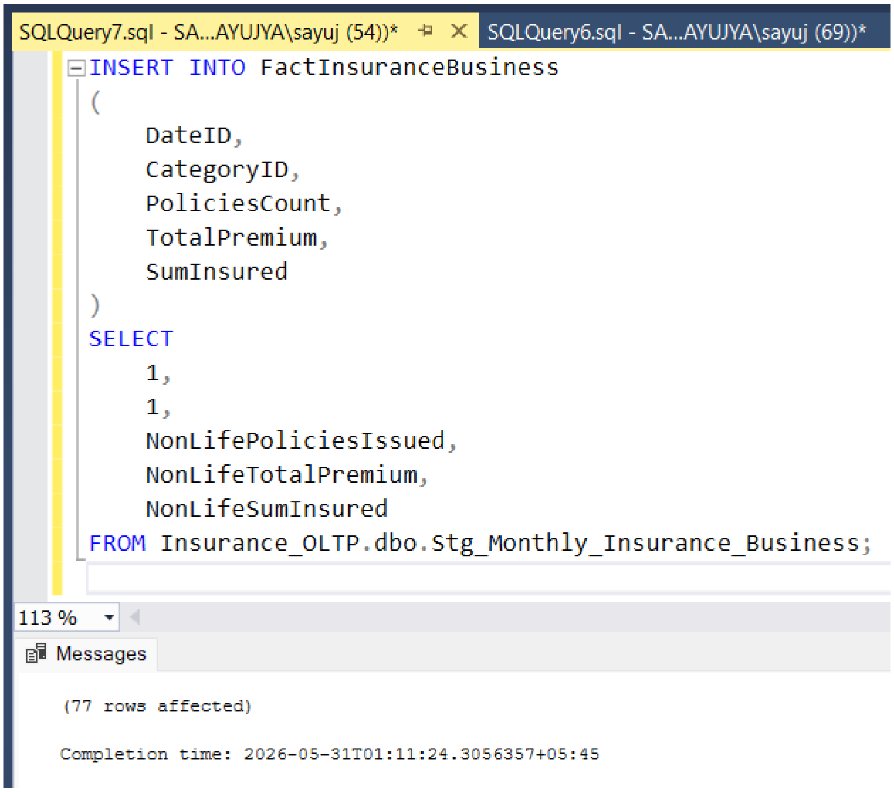


Figure 2.18 Successful loading of data into dw

Step 9: Verify Warehouse Data

Run queries to verify loaded records.

```
SELECT *
FROM FactInsuranceBusiness;
SELECT *
FROM DimInsuranceCategory;
SELECT *
FROM DimDate;
Screenshot 14: Data Warehouse Query Results
```

	FactID	DateID	CategoryID	PoliciesCount	TotalPremium	SumInsured
7	7	1	1	15142	1470.37	424778.92
8	8	1	1	4034	1600.08	433781.38
9	9	1	1	1824	388.64	3629473.14
10	10	1	1	2491	832.96	227223.62
11	11	1	1	26425	2894.58	729909.00
12	12	1	1	9615	1294.74	219018.13
13	13	1	1	29977	3909.64	1050576.41
14	14	1	1	18510	4330.33	559702.90
15	15	1	1	31306	6348.59	1612891.16

	CategoryID	CategoryName
1	1	Non-Life
2	2	Micro Non-Life

	DateID	MonthName	YearValue
1	1	Baishakh	2083

Figure 2.19 Final preview of the star schema

Result

The ETL process was successfully performed using Nepal Insurance Authority monthly insurance business data. Data was extracted from Excel files, cleaned and transformed in the staging area, and loaded into a dimensional data warehouse consisting of fact and dimension tables.

Conclusion

This lab provided practical experience in data warehousing and ETL processes using SQL Server. Raw insurance data from Excel files was imported into a staging table, transformed into a standardized format, and loaded into a star schema data warehouse. The exercise demonstrated how business data can be organized for efficient reporting, analysis, and decision-making in Business Intelligence systems.

Lab Work 3: Basic Reporting using SSRS

Objective

To create basic operational and analytical reports using SQL Server Reporting Services (SSRS) based on insurance data stored in the data warehouse.

Theory

- **SQL Server Reporting Services (SSRS):** SQL Server Reporting Services (SSRS) is a server-based reporting platform developed by Microsoft for creating, managing, and delivering reports. SSRS allows organizations to generate operational and analytical reports from various data sources and present them in tabular, graphical, or dashboard formats.
  - **Features of SSRS**
    - Report generation from SQL Server databases
    - Interactive and parameterized reports
    - Export to PDF, Excel, Word, and CSV formats
    - Web-based report access
    - Centralized report management

- **Report Server:** The Report Server is the core component of SSRS responsible for storing, processing, and delivering reports.
  - **Main Components**
    - Report Server Database
    - Web Service URL
    - Web Portal
    - Report Manager

The Report Server enables users to deploy and access reports through a web browser.

- **Tabular Reports:** A Tabular Report presents data in rows and columns similar to a spreadsheet.

- **Common Uses:**
  - Sales reports
  - Insurance business reports
  - Financial summaries
  - Performance reports
- **Advantages:**
  - Easy to read
  - Supports grouping and totals
  - Suitable for business reporting

Requirements

- SQL Server Express
- SQL Server Reporting Services (SSRS)
- SQL Server Data Tools (SSDT) or Visual Studio
- Insurance_DW Database
- Windows Operating System

Procedure

Step 1: Install SQL Server Reporting Services (SSRS)

- Download SQL Server Reporting Services (SSRS).
- Run the installer.
- Select Install Reporting Services.
- Complete the installation process.

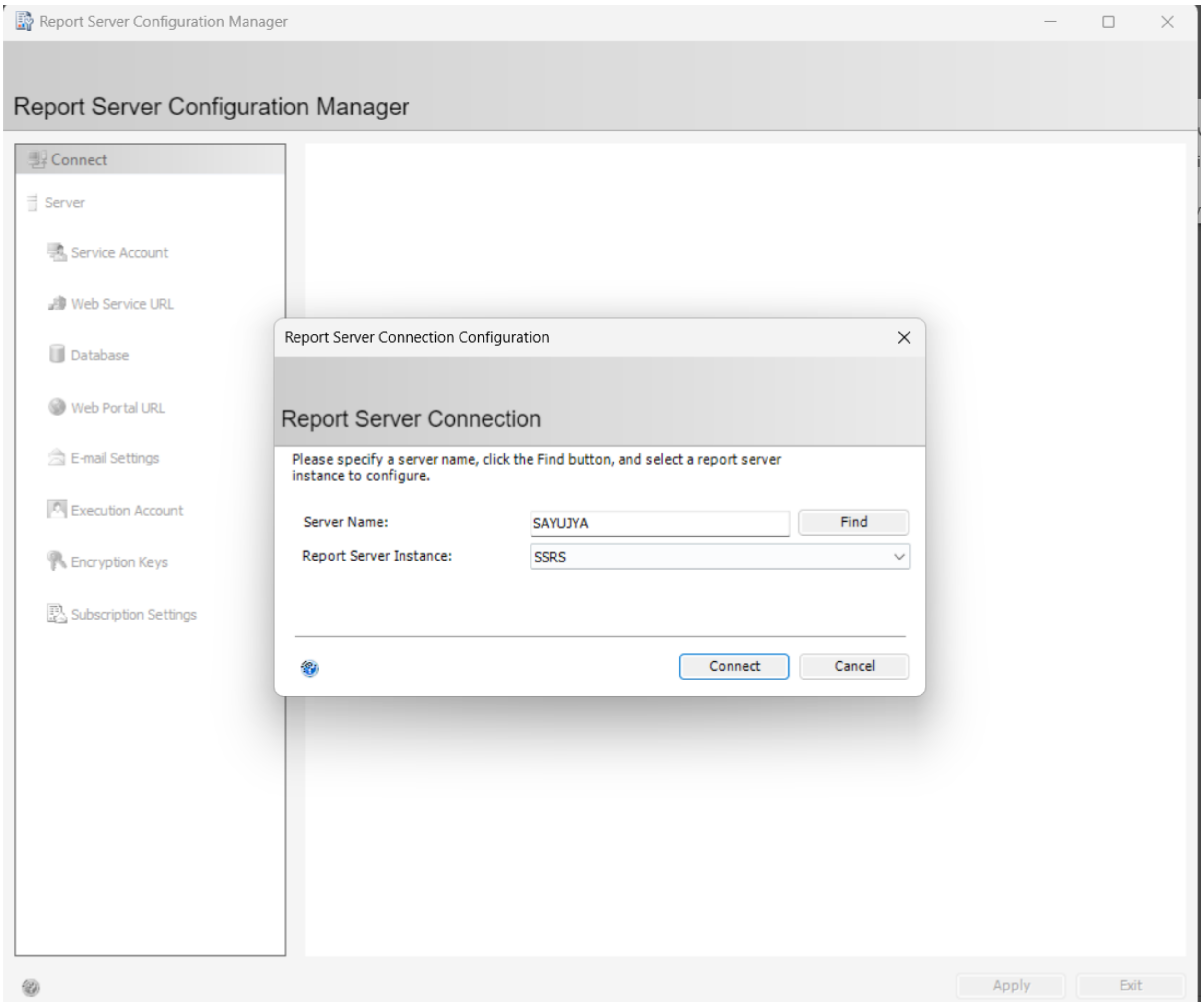


Figure 3.1 Successful Installation of SSRS

Step 2: Configure SSRS

- Open Report Server Configuration Manager.
- Connect to the SSRS instance.
- Configure:
    - Service Account
    - Web Service URL
    - Database
    - Web Portal URL
- Apply the configuration settings.

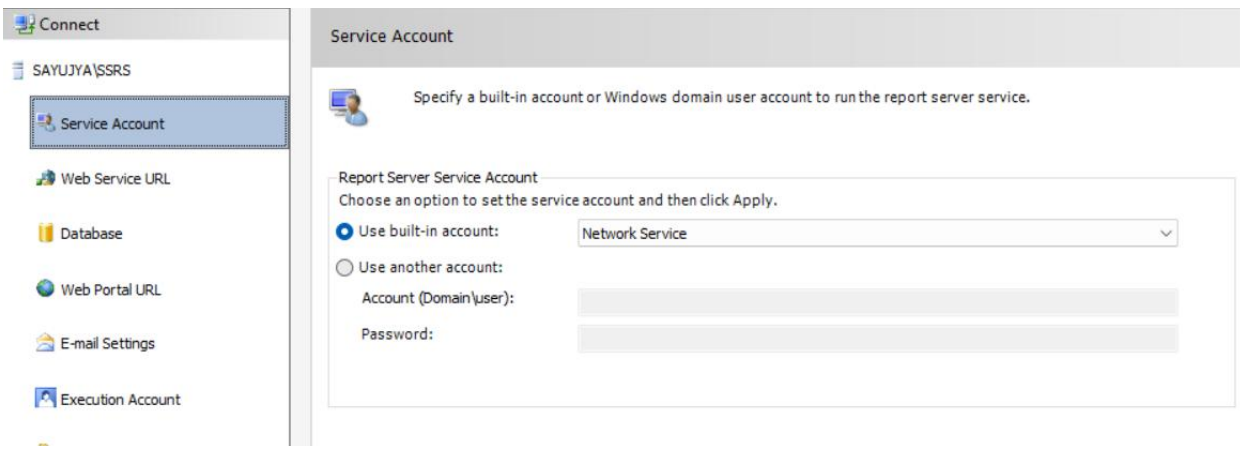


Figure 3.2 Service account configuration

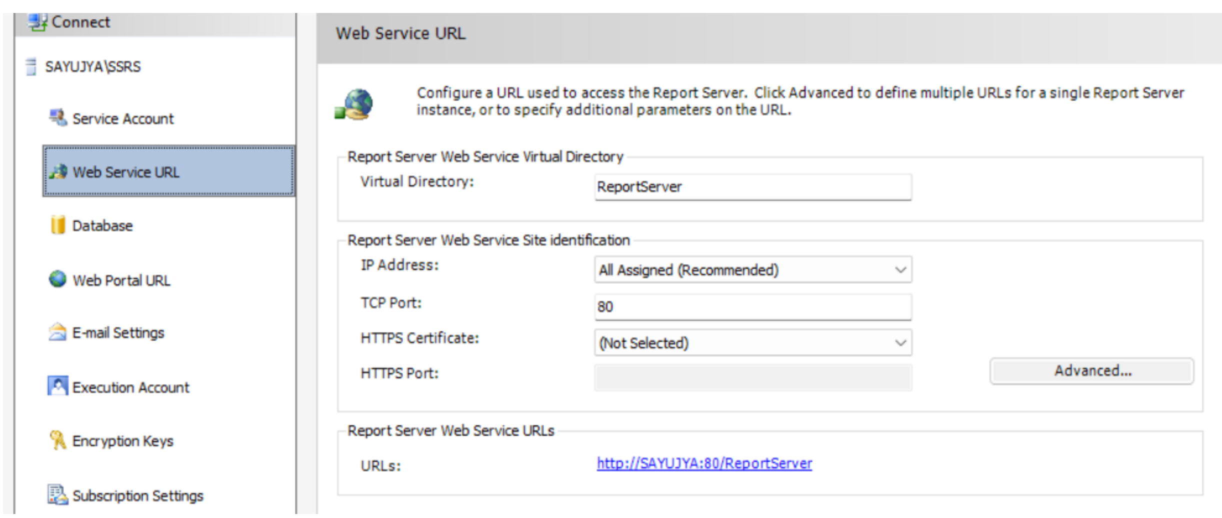


Figure 3.3 Web service configuration

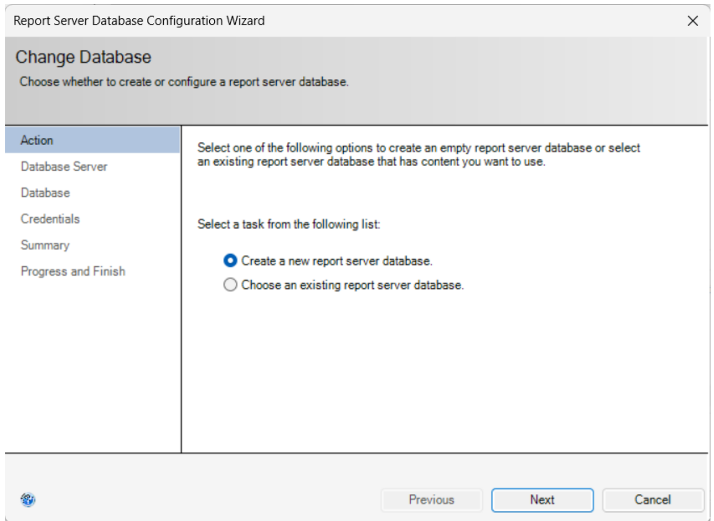


Figure 3.4 Creation of new database

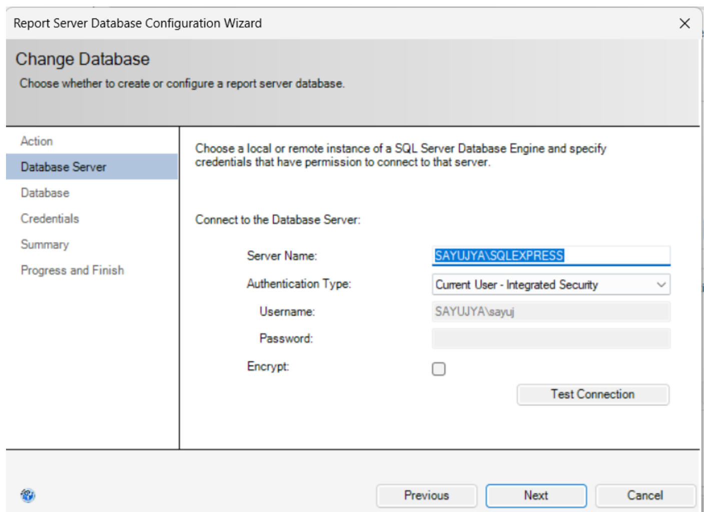


Figure 3.5 Connection to DB server

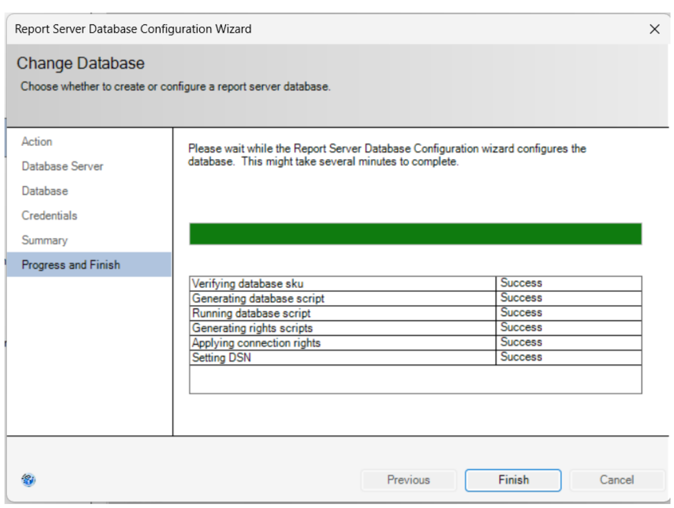


Figure 3.6 Finalize the new DB

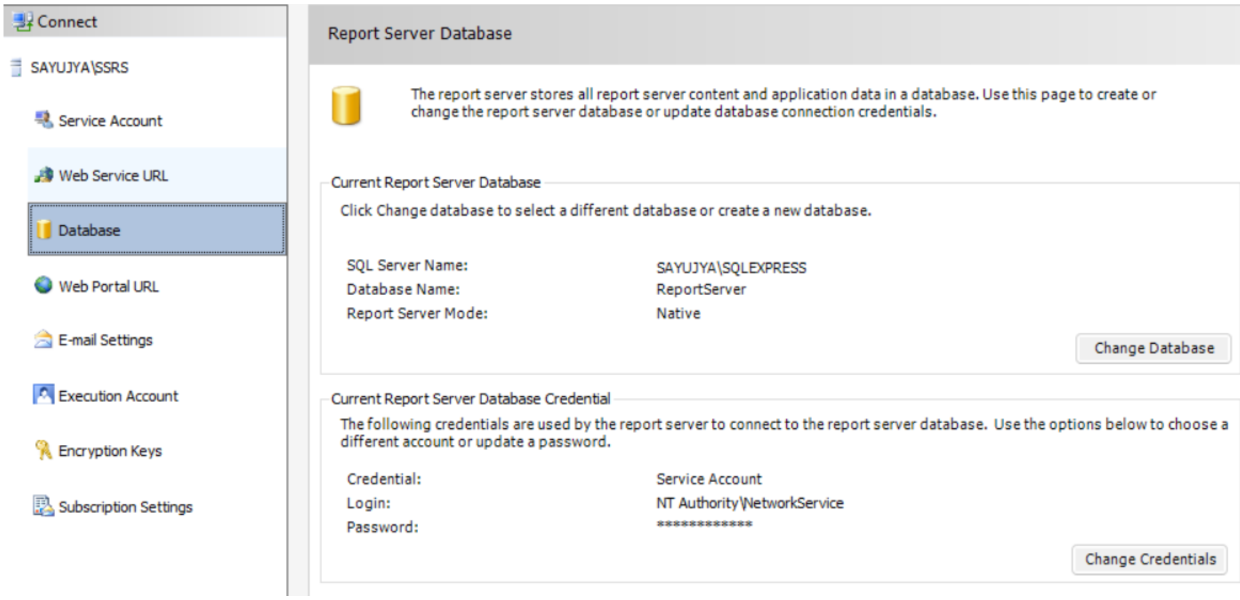


Figure 3.7 Final database configuration

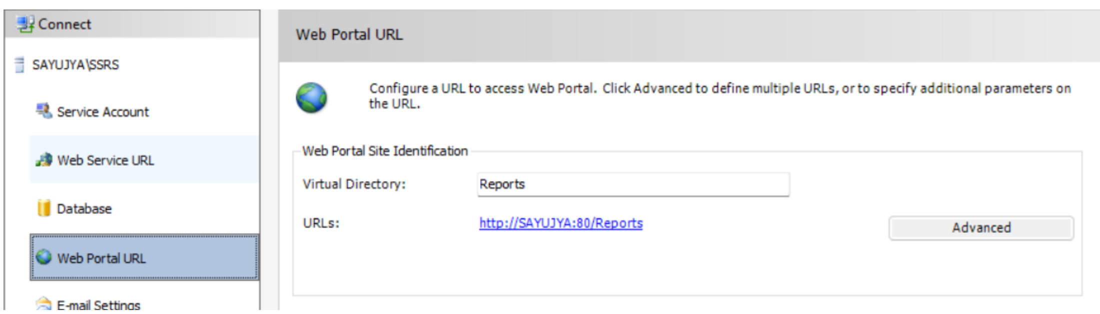


Figure 3.8 Webportal configuration

Step 3: Verify SSRS URLs

Verify the following URLs:

- **Web Service URL**

http://sayujya/ReportServer

Figure 3.9 Report server

- Web Portal URL

http://sayujya/Report

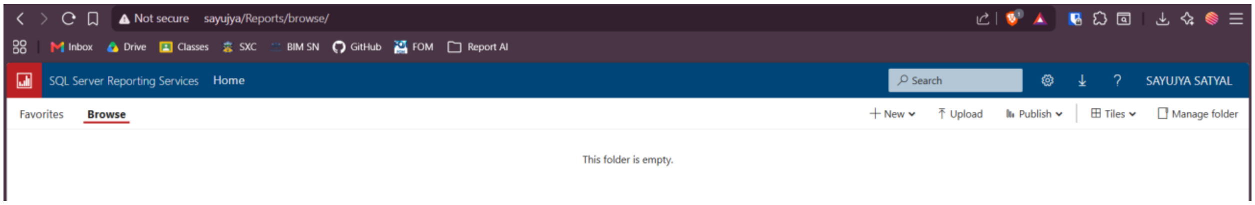


Figure 3.10 Web portal

Step 4: Open SQL Server Data Tools (SSDT)

- Open Visual Studio with SSDT installed.
- Create a new Report Server Project.
- Provide a project name such as: InsuranceReports
- Click Create.

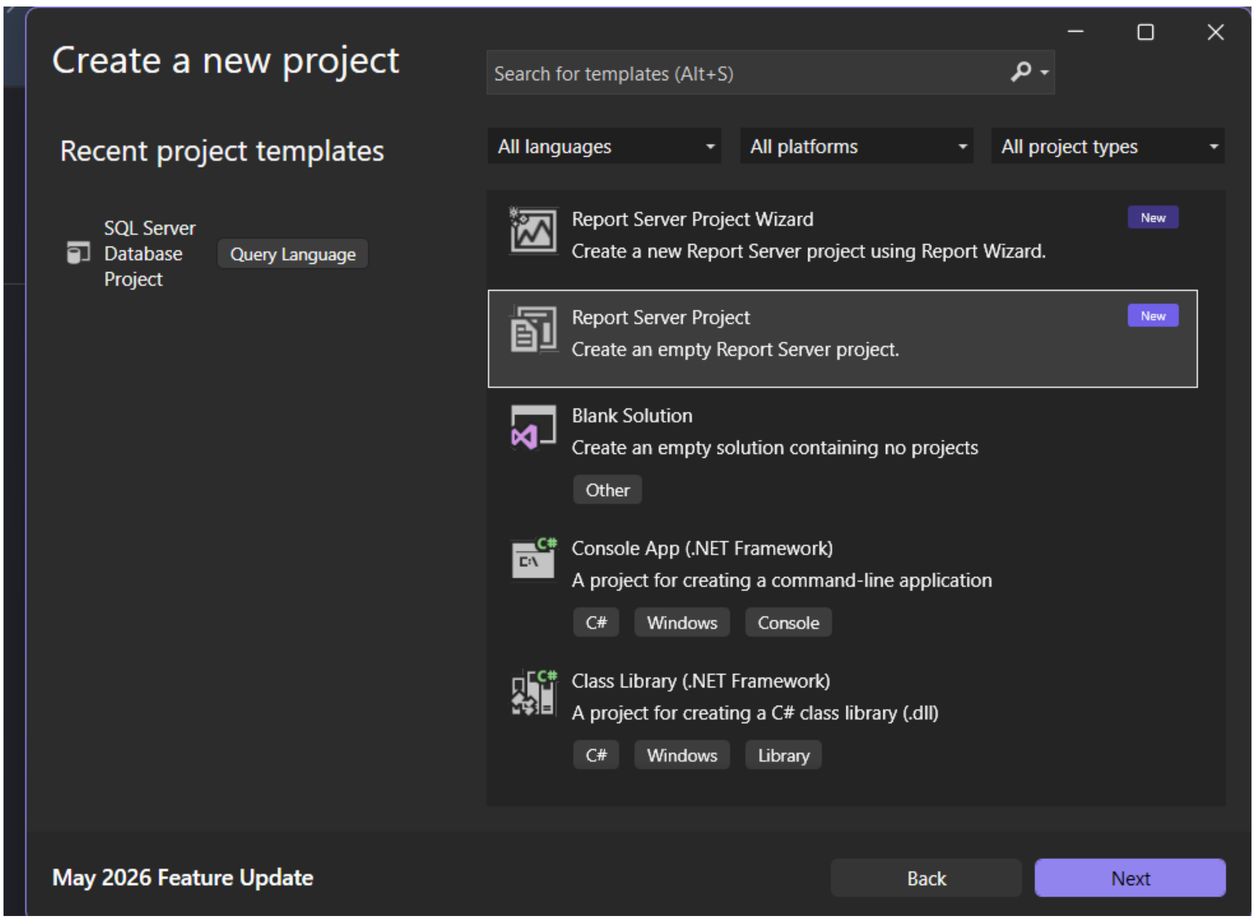


Figure 3.11 Creation of the project

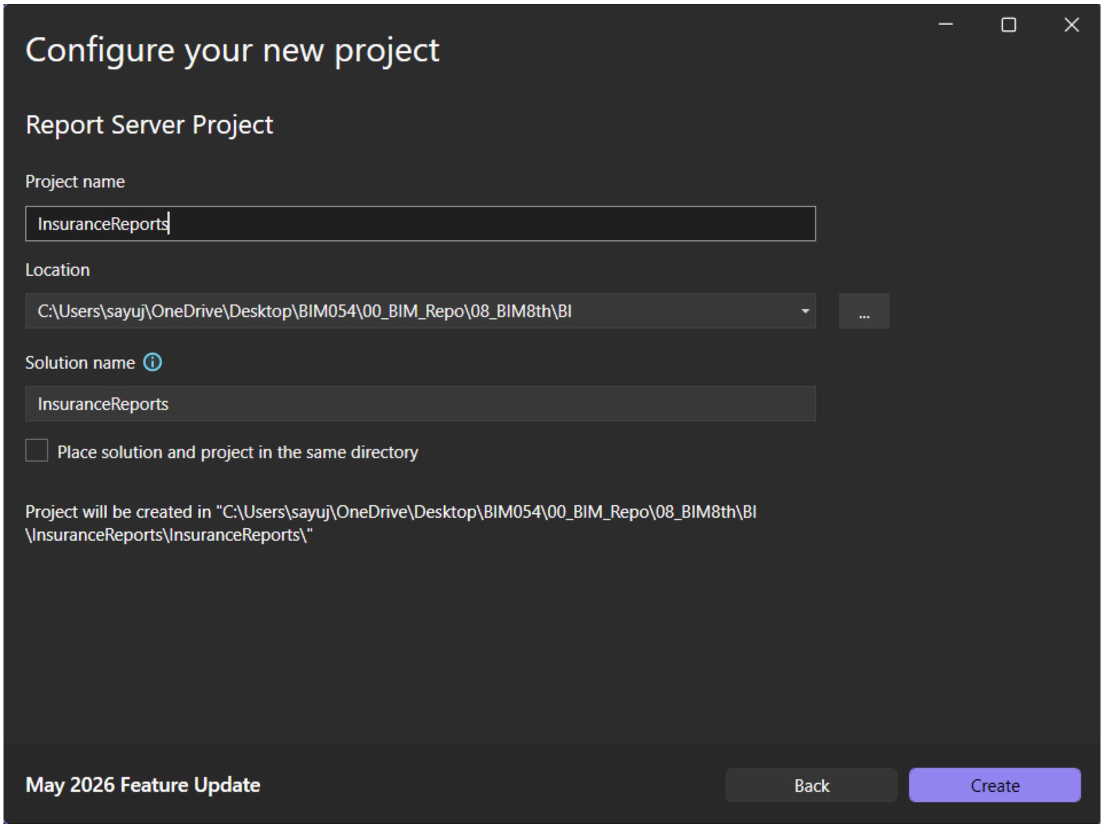


Figure 3.12 Configuration of new project

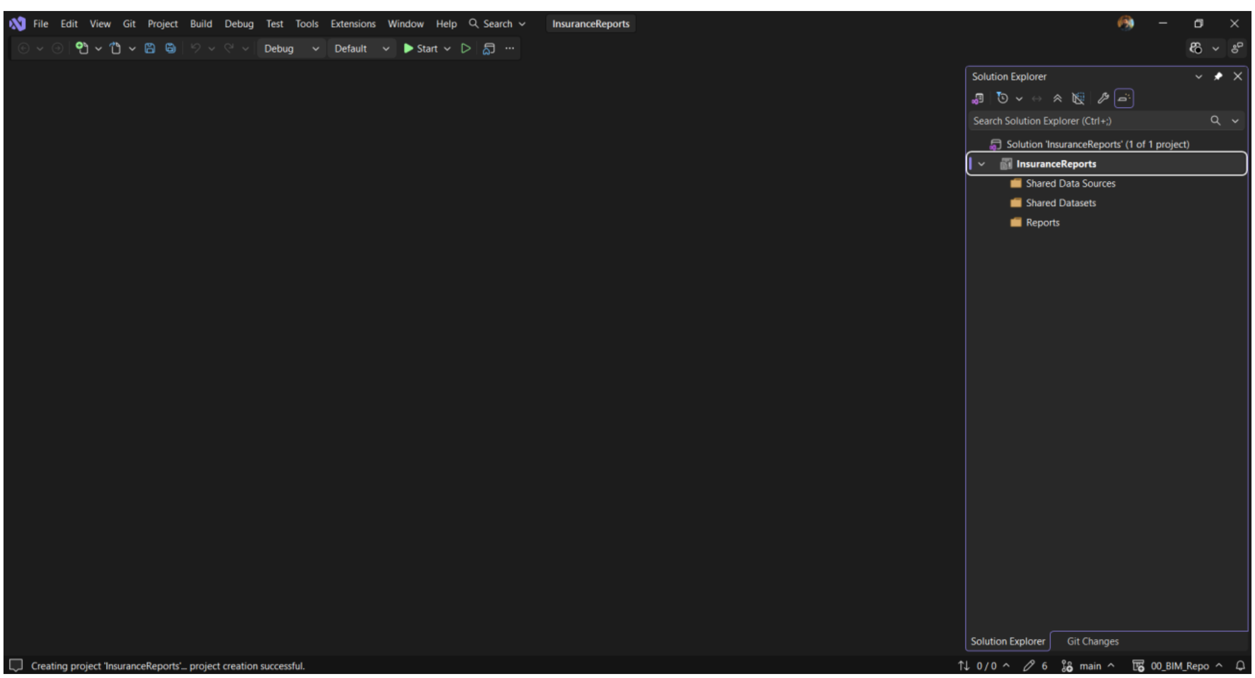


Figure 3.13 Successful Project Creation

Step 5: Create a Shared Data Source

- Right-click Shared Data Sources.
- Select Add New Data Source.
- Name the data source: InsuranceDW_DataSource and configure other settings

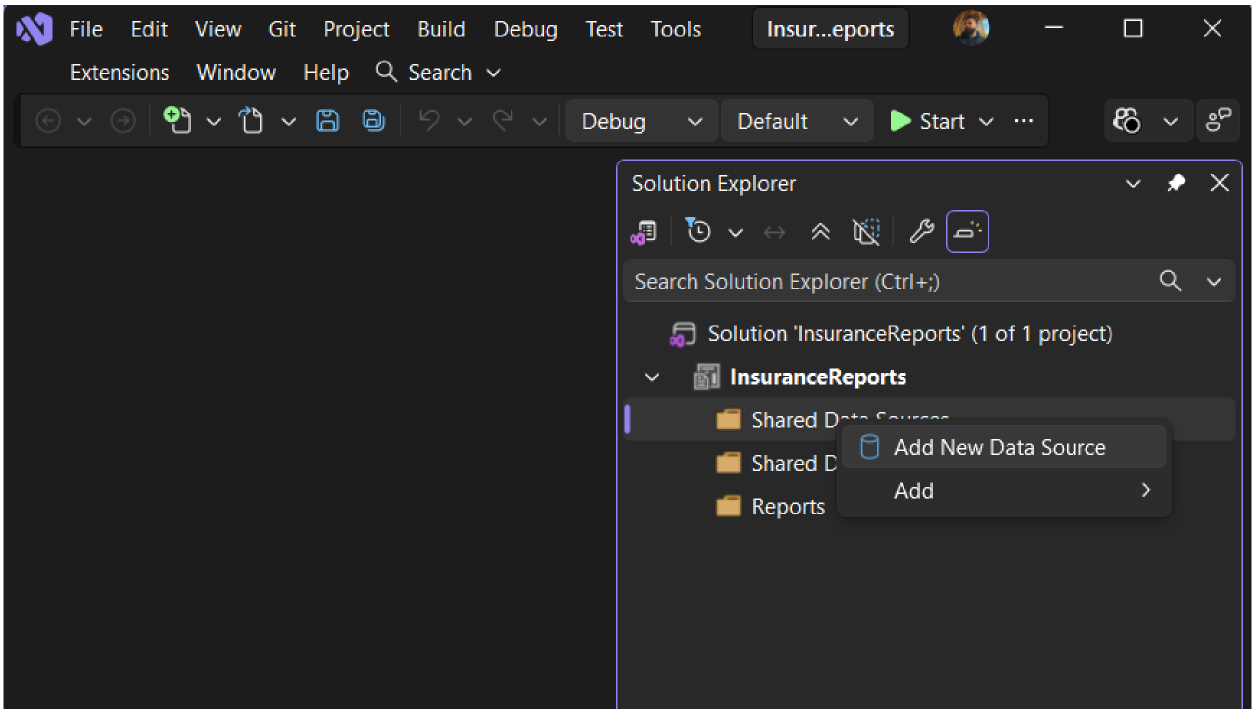


Figure 3.14 Adding new data source

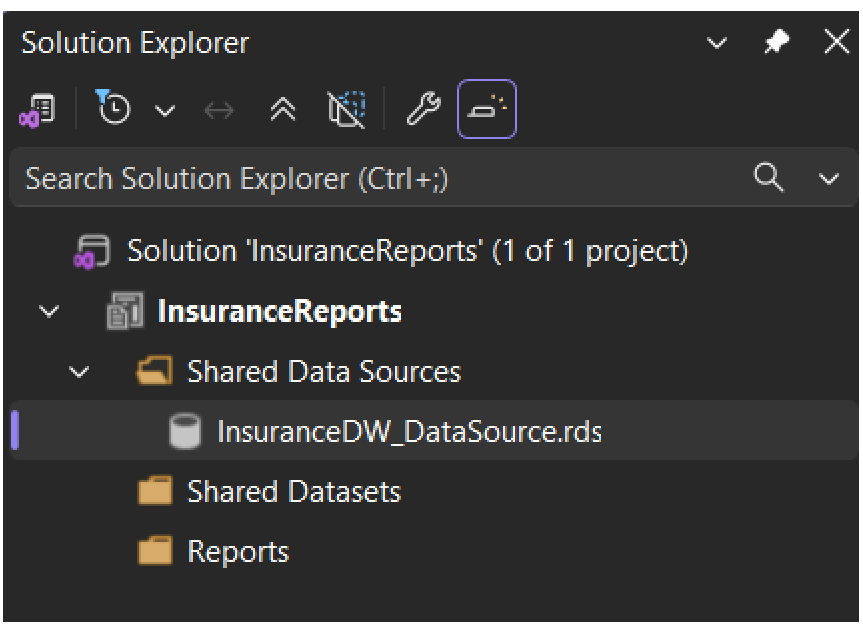


Figure 3.15 Successful creation of Shared Data sources

Step 6: Create a Dataset

- Create a dataset using the following SQL query:

```
SELECT
    d.MonthName,
    d.YearValue,
    c.CategoryName,
    f.PoliciesCount,
    f.TotalPremium,
    f.SumInsured
FROM FactInsuranceBusiness f
INNER JOIN DimDate d
    ON f.DateID = d.DateID
INNER JOIN DimInsuranceCategory c
    ON f.CategoryID = c.CategoryID;
```

- Execute the query and verify that data is returned.

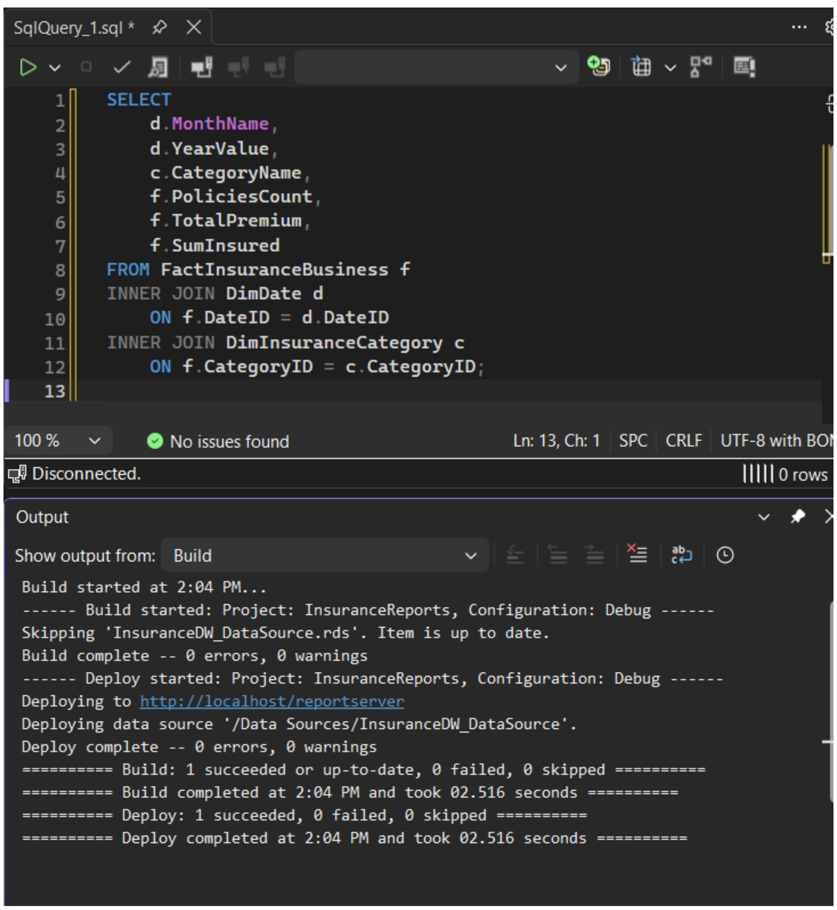


Figure 3.16 Successful query execution

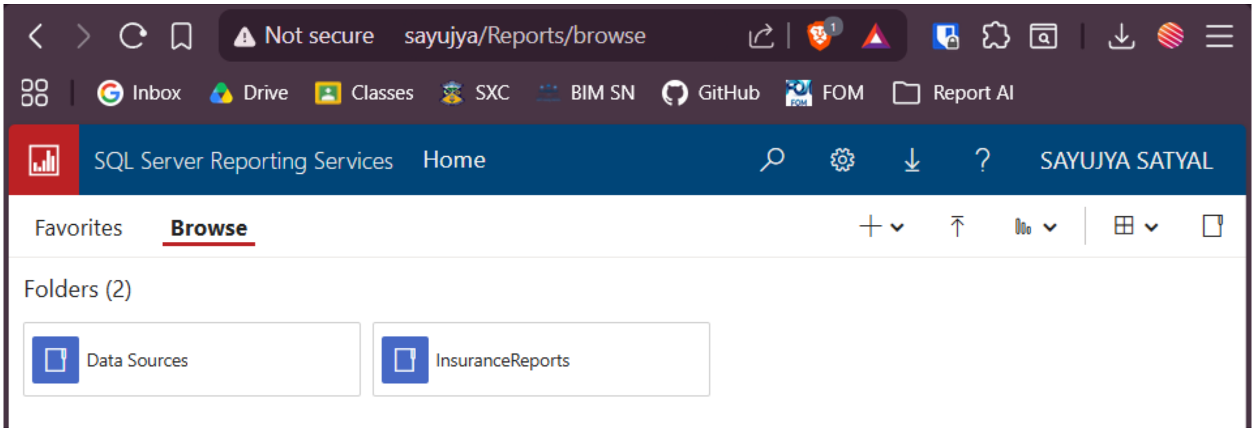


Figure 3.17 Creation of the new reports

Step 7: Design a Tabular Report

- Right-click Reports.
- Select Add New Report.
- Choose Table Wizard or Blank Report.
- Select the previously created dataset.
- Add the fields and group by category
- Generate the table layout.

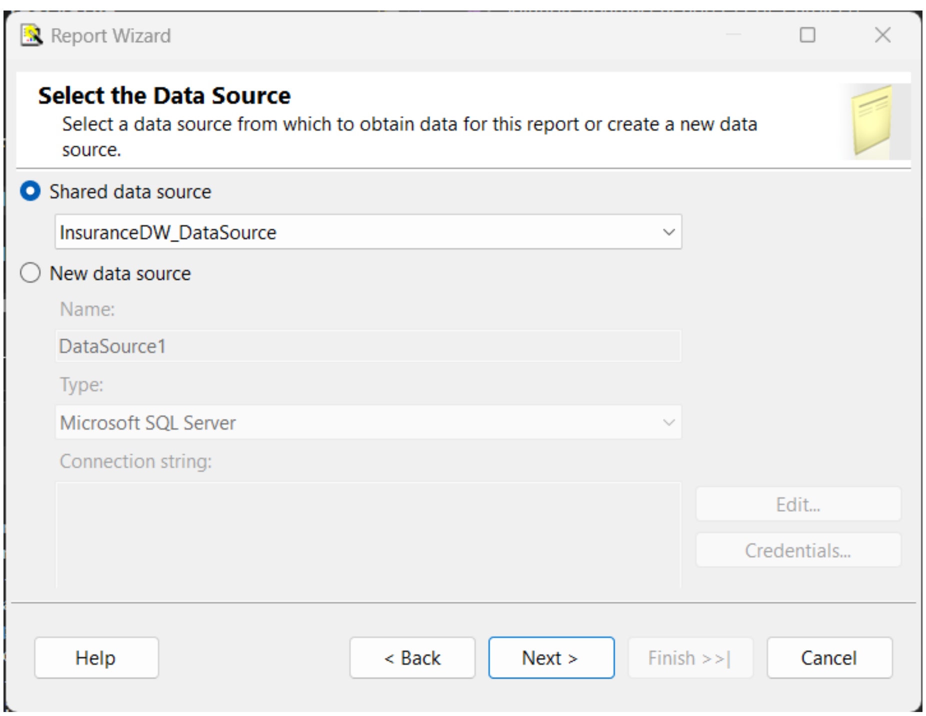


Figure 3.18 Selection of data source

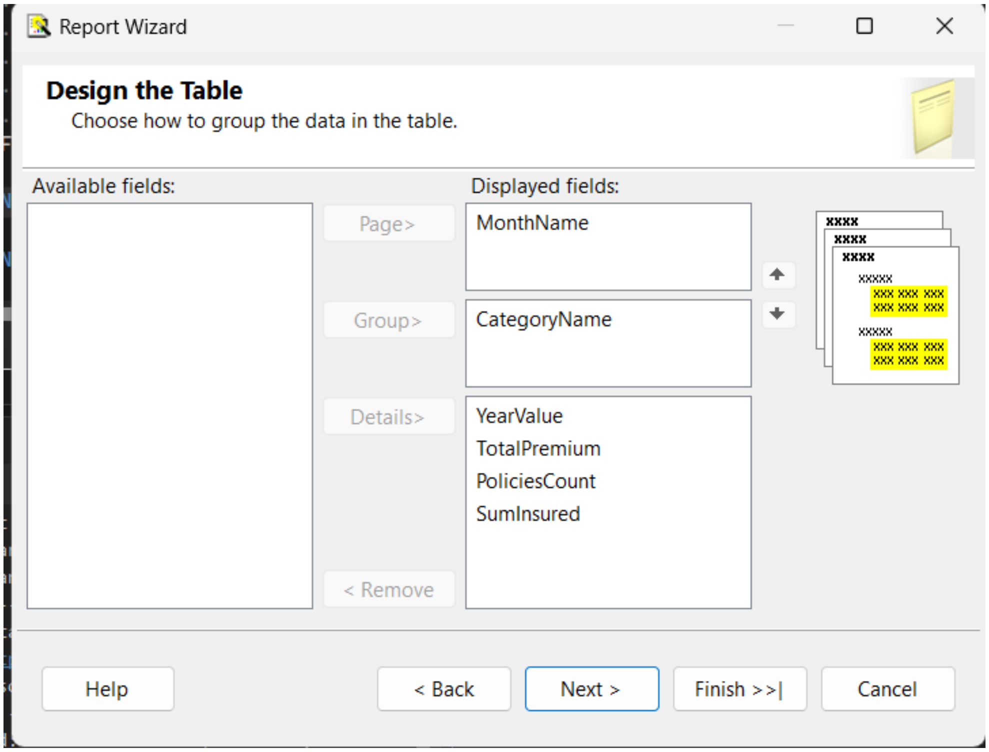


Figure 3.19 Designing the table

Step 8: Apply Formatting

- Perform the following formatting tasks:
    - Make headers bold
    - Apply background shading
    - Alignment
    - Center-align text fields
    - Right-align numeric values
    - Number Formatting
    - Premium Amount: Currency or Decimal Format
    - Policy Count: Integer Format
- Add total rows for:
    - Total Policies
    - Total Premium

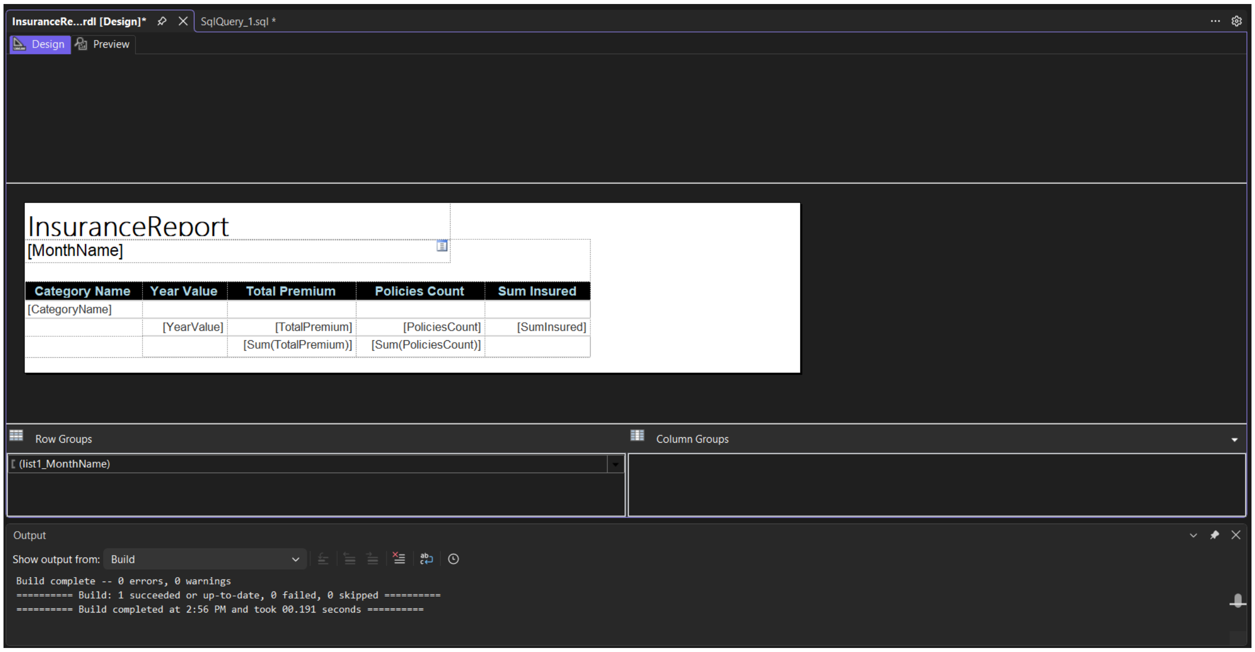
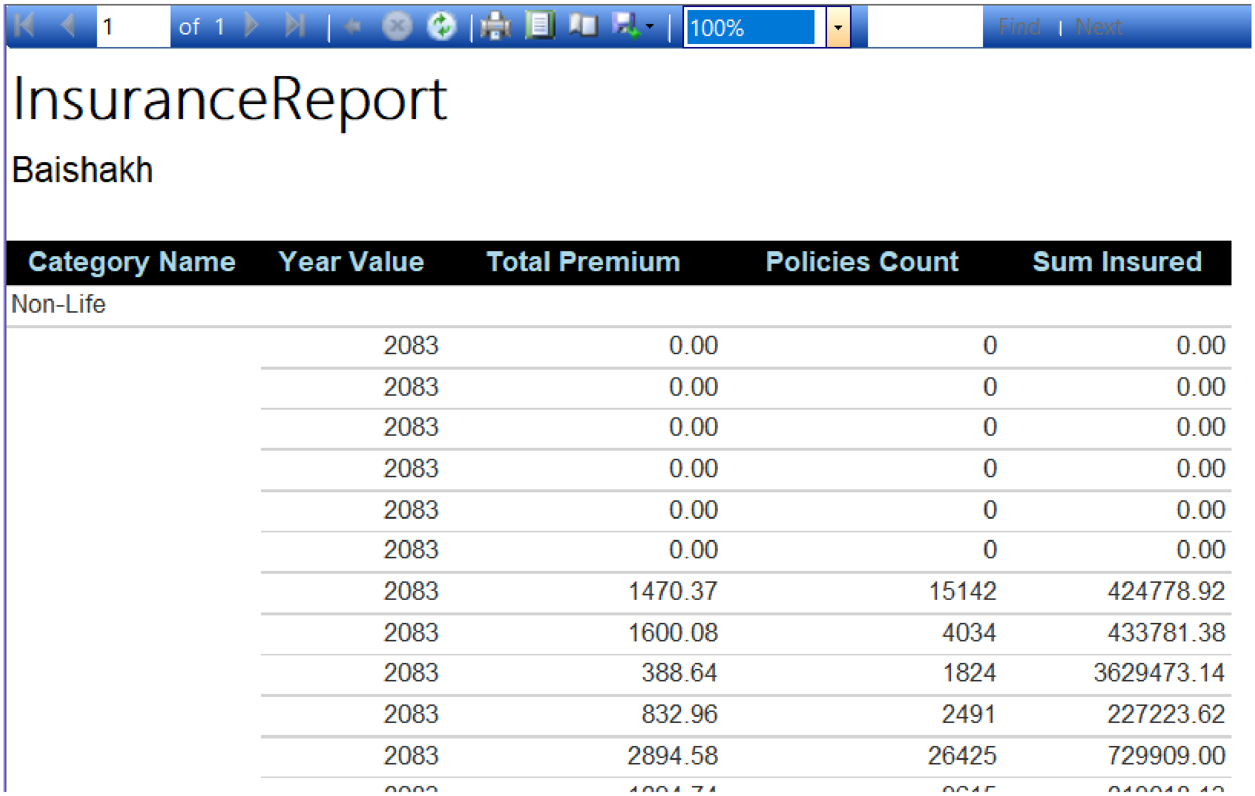


Figure 3.20 Designing the report

Step 9: Preview the Report

- Click the Preview tab.
- Verify:
    - Correct data retrieval
    - Proper grouping
    - Accurate totals
    - Correct formatting

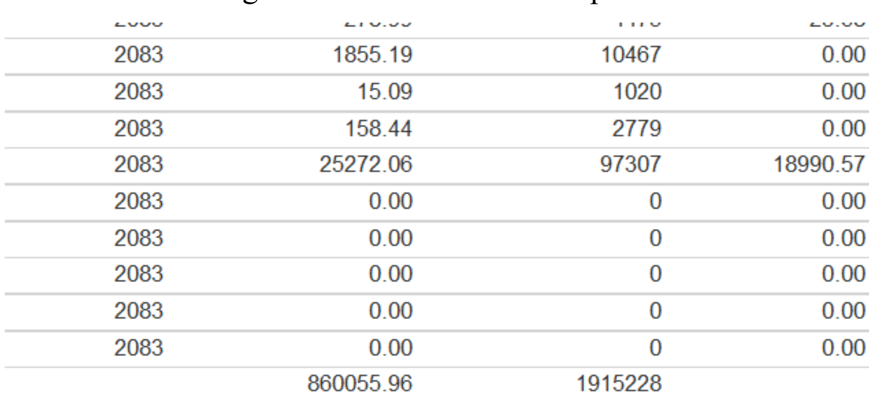


InsuranceReport

Baishakh

Category Name	Year Value	Total Premium	Policies Count	Sum Insured
Non-Life				
	2083	0.00	0	0.00
	2083	0.00	0	0.00
	2083	0.00	0	0.00
	2083	0.00	0	0.00
	2083	0.00	0	0.00
	2083	0.00	0	0.00
	2083	1470.37	15142	424778.92
	2083	1600.08	4034	433781.38
	2083	388.64	1824	3629473.14
	2083	832.96	2491	227223.62
	2083	2894.58	26425	729909.00

Figure 3.21 Preview of the report



2083	1855.19	10467	0.00
2083	15.09	1020	0.00
2083	158.44	2779	0.00
2083	25272.06	97307	18990.57
2083	0.00	0	0.00
2083	0.00	0	0.00
2083	0.00	0	0.00
2083	0.00	0	0.00
2083	0.00	0	0.00
	860055.96	1915228	

Figure 3.22 Total of each of the columns

Step 11: Deploy the Report

- Open Project Properties.
- Select Deploy.
- Wait for the deployment to complete successfully.

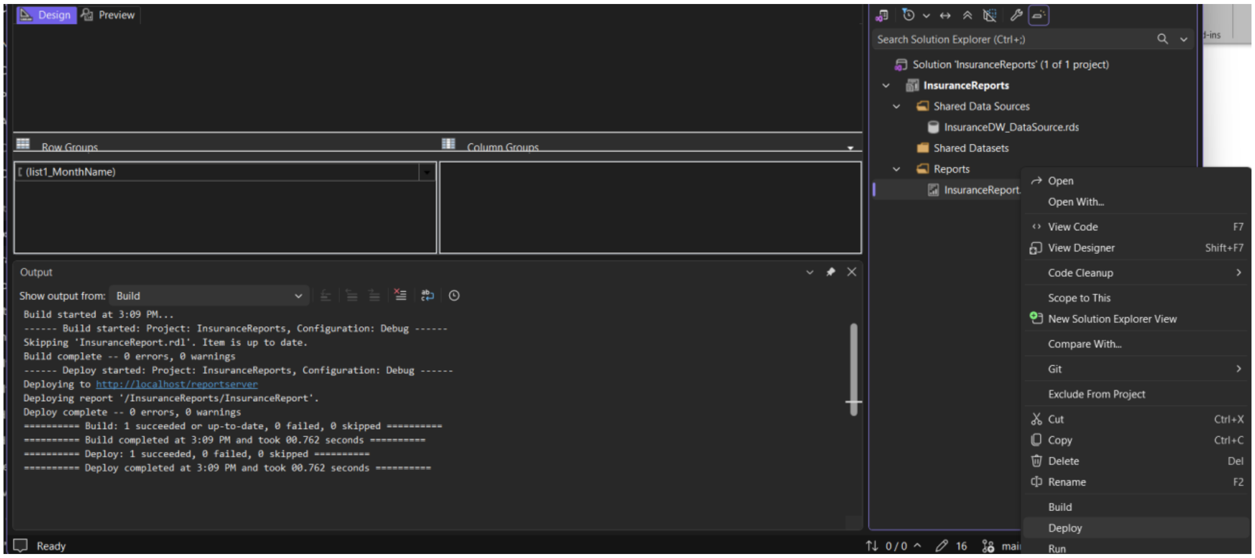


Figure 3.23 Successful deployment of the report

Step 12: Access the Report through SSRS Web Portal

- Open the SSRS Web Portal.
- Locate the deployed report.
- Open and run the report.

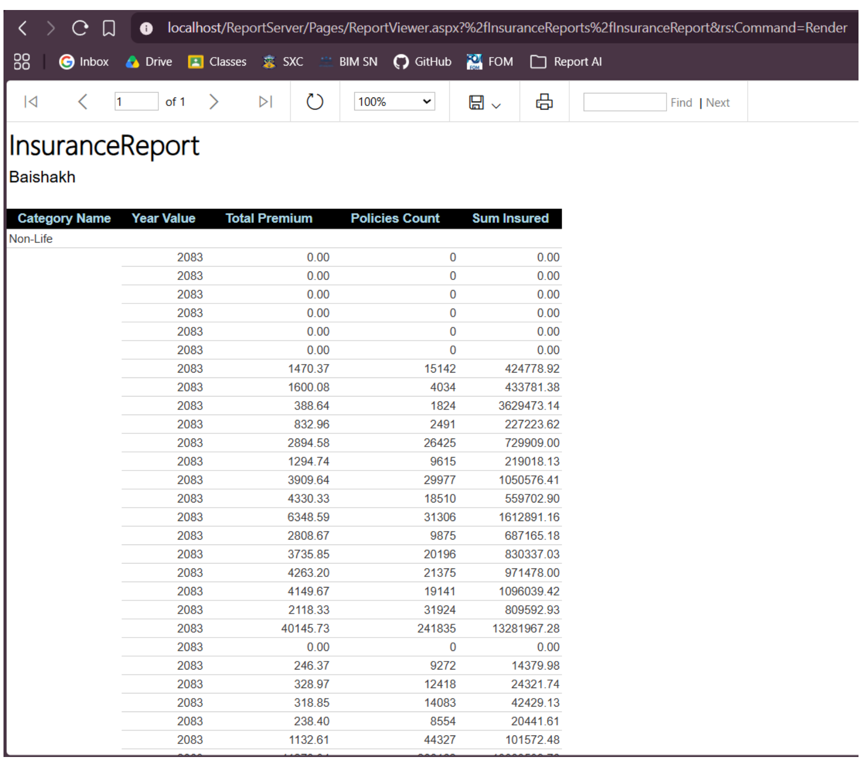


Figure 3.24 Verify the report through SSRS portal

Result

A tabular insurance business report was successfully created using SQL Server Reporting Services (SSRS). The report retrieved data from the Insurance_DW data warehouse, displayed insurance business metrics, applied grouping and formatting, and was successfully deployed to the Report Server.

Conclusion

This lab demonstrated the process of creating business reports using SSRS. A shared data source and dataset were created, a tabular report was designed using warehouse data, grouping and formatting were applied, and the final report was deployed to the Report Server. The exercise provided practical experience in business reporting and report deployment using SQL Server Reporting Services.